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IMPROVING LONG-RANGE LIDAR MEASUREMENT ACCURACY THROUGH ACTIVE GIMBAL STABILISATION IN DYNAMIC ENVIRONMENTS

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Abstract

Unmanned Ground Vehicles (UGVs) operating in uneven outdoor terrain are subjected to terrain-induced roll and pitch motion that directly affects the orientation and stability of onboard perception sensors. For rotation Light Detection and Ranging (LiDAR) systems, residual sensor tilt during scan acquisition introduces range-dependent geometric distortion and misalignment between successive point clouds, reducing the reliability of mapping, localisation, and autonomous navigation. Although active gimbal stabilisation has been widely adopted in aerial and maritime applications, its quantitative impact on LiDAR geometric consistency for ground-based UGVs remains insufficiently investigated, particularly for long-range perception where small residual attitude errors propagate into significant geometric distortion.

This thesis presents the design and simulation-based evaluation of an actively stabilised Two Degrees of Freedom (2-DOF) platform for the Swedish Land-based Robotics Centre (SLaRC) UGV. The proposed platform employs an Inertial Measurement Unit (IMU)-driven gravity-aligned stabilisation architecture to actively compensate for terrain-induced roll and pitch disturbances. Evaluation was performed using experimentally recorded UGV attitude data acquired during real terrain traversal, which was applied as disturbance input to a closed-loop Simscape Multibody simulation model. Stabilisation performance was evaluated using angular Root Mean Square (RMS) error, Disturbance Rejection Ratio (DRR), frequency-domain analysis, and point cloud based geometric comparison metrics.

The simulated stabilisation platform achieved a roll DRR of $59.9\times$ *and a pitch DRR of* $7.8\times$ *under terrain-induced disturbances with peak chassis excursions of* $+31^\circ$ *on the roll axis and* $\pm 11.9^\circ$ *on the pitch axis. Relative to a rigidly mounted baseline configuration, the stabilised platform reduced mean Cloud-to-Cloud (C2C) distance by* $15.6\times$ *and reduced Iterative Closest Point (ICP) registration error by* $4.0\times$*. Furthermore, the stabilised configuration substantially reduced geometric distortion and improved point cloud consistency in both C2C and Multiscale Model to Model Cloud Comparison (M3C2) analyses.*

The results demonstrate that active roll-pitch stabilisation can significantly improve LiDAR geometric consistency during uneven terrain traversal. The work further establishes a quantitative evaluation framework linking residual stabilisation error to perception-relevant geometric accuracy for ground-based mobile robotic platforms.

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List of Abbreviation

2-DOF Two Degrees of Freedom

3-DOF Three Degrees of Freedom

AI Artificial Intelligence

AHRS Attitude and Heading Reference System

C2C Cloud-to-Cloud

CAN Controller Area Network

CAN-FD Controller Area Network with Flexible Data-Rate

CSV Comma-Separated Values

DOF Degrees of Freedom

DRR Disturbance Rejection Ratio

FFT Fast-Fourier Transform

FIR Finite Impulse Response

GPS Global Positioning System

ICP Iterative Closest Point

IMU Inertial Measurement Unit

INS Inertial Navigation System

LiDAR Light Detection and Ranging

LQR Linear-quadratic Regulator

M3C2 Multiscale Model to Model Cloud Comparison

MEMS Micro Electro Mechanical Systems

RANSAC Random Sample Consensus

PID Proportional Integral Derivative

PSD Power Spectral Density

RMS Root Mean Square

ROS2 Robot Operating System 2

SR Set-Reset

SLAM Simultaneous Localization and Mapping

SLaRC Swedish Land-based Robotics Centre

SMC Sliding Mode Control

UAV Unmanned Aerial Vehicle

UGV Unmanned Ground Vehicle

WBS Work Breakdown Structure

1. Introduction

Unmanned Ground Vehicles (UGVs) are increasingly deployed in outdoor and challenging terrain, for example, dense forests, for applications such as search and rescue, infrastructure inspection, and autonomous exploration [1], [2], [3], [4]. Their value has grown due to their ability to operate in environments that are dangerous or otherwise inaccessible to humans. Their use has also expanded into civilian and commercial domains, including autonomous delivery systems, agricultural robotics, and service robots operating in semi-structured outdoor environments. Reliable perception is a fundamental requirement for UGVs operating in teleoperated, semi-autonomous, and fully autonomous modes. Sensor modalities such as LiDARs and cameras provide the environmental information required for mapping, localisation, and navigation [5]. Unlike structured indoor environments, outdoor terrain introduces significant disturbances caused by uneven terrain surfaces, which induce roll and pitch motion during vehicle traversal [6].

In many commercial UGV systems, perception sensors are rigidly mounted to the vehicle chassis.[1] While suspension systems and passive damping mechanisms can attenuate high-frequency vibrations, they do not compensate for sustained orientational changes caused during terrain traversal [7], [8]. The SLaRC UGV used as a baseline platform in this thesis currently has rigidly mounted sensors. As a result, platform motion is directly transferred to the sensor frame of the UGV, degrading the consistency of acquired sensor measurements. Degraded perception quality for vision-based sensors can manifest as motion blur, inconsistent viewpoints, and reduced feature tracking performance. For LiDAR sensors, platform motion during scan acquisition can introduce motion distortions and misalignment between successive point clouds [9], [10]. These effects reduce the robustness of perception pipelines and limit the reliability of autonomous operation in challenging terrain.

Active sensor stabilisation using gimbal mechanisms offers a means to decouple sensor orientation from the motion of the UGV by actively compensating for roll and pitch disturbances caused during terrain traversal [11]. By maintaining a near-horizontal sensor orientation relative to gravity, active gimbals can improve spatial consistency and temporal coherence of sensor data, thereby enhancing the performance of perception and navigation. While such systems are commonly employed in aerial platforms and camera stabilisation systems, their integration on UGVs presents unique challenges due to increased vibration levels and sudden mechanical shocks [11], [12].

This thesis investigates the design and simulation-based evaluation of an actively stabilised platform for the lightweight SLaRC UGV, intended for uneven terrain traversal. The work focuses on the development of an IMU-driven gravity-aligned roll- and pitch-compensating gimbal architecture and evaluates its ability to improve sensor orientation stability relative to a rigid chassis-mounted baseline. The evaluation emphasises disturbance rejection performance and residual attitude error under terrain-induced roll and pitch motion up to approximately $\pm 30°$, corresponding to the operational limits of the SLaRC platform.

Despite extensive research on gimbal stabilisation, control strategies, and point cloud processing, few studies provide a direct quantitative link between stabilisation performance and LiDAR point cloud quality under realistic terrain-induced disturbances. In particular, the relationship between platform attitude dynamics and the resulting geometric distortion in point clouds remains insufficiently quantified in the literature.

To address this research gap, this thesis evaluates both stabilisation performance and its direct impact on LiDAR data quality using both stabilisation metrics and perception-relevant geometric metrics.

[1] Examples include the Robotnik SUMMIT (https://robotnik.eu/products/mobile-robots/rb-summit/), RoboDyne MAXXII Standard Tracked Robot (https://www.robo-dyne.com/maxxll/), and AgileX SCOUT 2.0 UGV Chassis Platform (https://global.agilex.ai/products/scout-2-0).

2. Problem Formulation and Research Objectives

As established in Section 1., terrain-induced roll and pitch motion degrades LiDAR-based perception on UGVs when sensors are rigidly mounted to the vehicle chassis. Although the existing suspension system attenuates high-frequency vibrations, it does not decouple the sensor frame from low-frequency attitude variations caused by terrain geometry. As a result, LiDAR measurements are acquired under continuously varying sensor orientations, reducing point-cloud consistency and limiting overall perception reliability.

Active sensor stabilisation using gimbal mechanisms has been proposed as a means to mitigate these effects by decoupling the sensor orientation from the vehicle motion. Although such systems are established in aerial and maritime domains, their vehicle-integrated performance in ground-based UGVs applications remains insufficiently quantified. In particular, there is limited experimental evidence directly linking stabilisation performance to measurable improvements in LiDAR geometric consistency under realistic terrain-induced disturbances.

Even with active stabilisation, residual orientation error may persist during dynamic traversal. For long-range LiDAR sensing, small angular deviations propagate into range-dependent geometric error, resulting in vertical and positional distortions within reconstructed point clouds. Therefore, stabilisation performance must be evaluated not only in terms of angular disturbance rejection, but also in relation to perception-relevant geometric accuracy. An experimentally grounded evaluation is therefore required to determine whether active roll-pitch stabilisation provides sufficient improvement in LiDAR data integrity to justify the added mechanical, computational, and control complexity compared to rigid mounting. To quantitatively assess active LiDAR stabilisation on ground-based platforms, this thesis investigates the design and simulation-based evaluation of a 2-DOF stabilised sensor platform mounted on the SLaRC UGV, illustrated in Figure 4. Yaw stabilisation was excluded since the research questions focused specifically on roll and pitch as the dominant contributors to gravity-referenced LiDAR pointing error during uneven terrain traversal. The work is guided by the following research questions:

RQ1: What level of RMS roll and pitch attenuation can be achieved by a 2-DOF actively stabilised platform under terrain-induced disturbances up to $\pm 30°$, corresponding to the operational limits of the SLaRC UGV.

RQ2: To what extent can active stabilisation enable continuous LiDAR data acquisition during UGV motion?

RQ3: Does active 2-DOF stabilisation produce a measurable improvement in LiDAR geometric consistency under terrain-induced disturbances, relative to a rigid mounting baseline?

2.1 Thesis Contributions

This thesis makes the following contributions:

1. A simulation-based vehicle-integrated evaluation framework for active roll and pitch stabilisation of a roof-mounted LiDAR platform on a ground-based UGV.
2. A quantitative derivation of stabilisation accuracy requirements based on allowable LiDAR geometric distortion at long measurement ranges.
3. A perception-oriented evaluation framework that links angular stabilisation metrics (RMS error, DRR) to perception-relevant geometric deviation.
4. A quantitative comparative evaluation between rigidly mounted and actively stabilised LiDAR configurations under representative terrain-induced disturbances.

3. Background

UGVs are ground-based mobile robotics platforms designed to operate without an onboard human operator, either through teleoperation or autonomous control [5], [13]. UGVs have been extensively studied and deployed in different sectors, primarily in military and defense-related contexts, where they have been used for tasks such as reconnaissance, logistics, and material transport in hazardous environments, as well as civilian applications [4], [13].

Due to limitations in onboard sensing, localization, and computational resources, early UGV systems relied on teleoperation rather than autonomous navigation. As sensing technologies and embedded computing capabilities improved, UGV platforms were increasingly deployed in civilian applications such as search and rescue and infrastructure inspection [14]. These applications typically involve operation in outdoor environments and challenging terrain, where uneven terrain induces roll and pitch motion of the UGV platform. Platform motion has been shown to affect both sensor orientation and the geometric consistency of acquired sensor data for rigidly mounted sensor modalities, making it an important consideration in UGV design [5], [15]. Experimental studies on off-road UGV platforms report terrain-induced roll and pitch motion of ± 10 degrees under realistic outdoor operating conditions [16], [17]. The vehicle-fixed coordinate frame and the corresponding roll, pitch, and yaw rotations are illustrated in Figure 1.

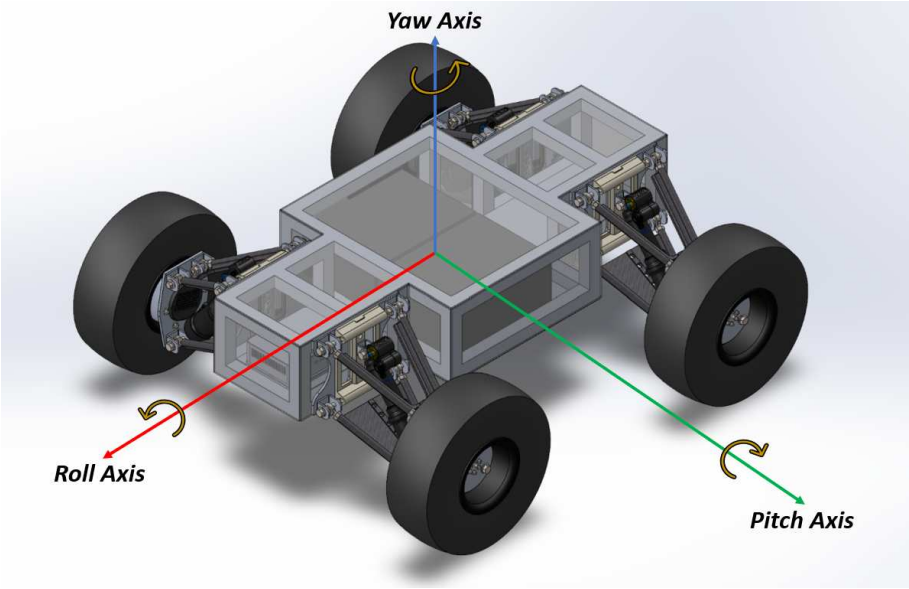


Figure 1: Definition of roll, pitch, and yaw axes with respect to the body-fixed coordinate frame of the UGV. The roll, pitch, and yaw motions correspond to rotations about the body-fixed x-, y-, and z-axes, respectively.

3.1 Terrain Traversal and Sensor Degradation

In the context of UGVs, platform motion is dominated by terrain-induced roll and pitch variations rather than smooth, continuous trajectories. Uneven ground, negative obstacles, and slopes introduce rapid and frequent attitude changes that propagate directly to rigidly mounted sensor platforms [11]. Such disturbances may also degrade vision-based perception through inconsistent viewpoints and motion-induced image artefacts [7]. For rotating LiDAR sensors, platform motion during scan acquisition can introduce geometric distortions, scan-line misalignment, and inconsistencies between successive point clouds [8]. Furthermore, small angular deviations in sensor orientation can introduce significant geometric distortion in LiDAR point clouds, particularly at longer sensing ranges where minor pitch and roll variations propagate into large geometric discrepancies in the reconstructed environment [18].

Experimental studies have shown that platform motion can introduce systematic geometric

distortion in mobile mapping systems when sensor attitude is insufficiently compensated [19], [20]. While suspension systems and passive damping mechanisms are present in the SLaRC UGV, they do not compensate for sustained orientation changes caused by terrain geometry. Consequently, rigidly mounted sensor platforms remain fundamentally coupled to vehicle attitude during uneven terrain traversal. Experimental studies of off-road UGVs further indicate that terrain-induced roll and pitch disturbances are predominantly concentrated at low frequencies during uneven terrain traversal [21]. Consequently, stabilisation performance depends not only on static orientation accuracy, but also on the dynamic disturbance rejection capability of the stabilisation system.

3.2 LiDAR and Point Clouds

LiDAR is an active remote sensing technology that measures distances by emitting laser pulses and recording the time-of-flight of the reflected signal [22], [23], [24], [25]. By continuously scanning the environment, LiDAR systems generate three-dimensional representations of surrounding surfaces known as point clouds. These point clouds consist of discrete 3D points representing measured surface locations within the environment. The resulting point clouds provide geometric information used for mapping, localisation, and environment reconstruction. LiDAR is widely used in robotics due to its high accuracy and rapid three-dimensional environment acquisition. Mobile LiDAR systems acquire point cloud data from moving platforms and are widely used in autonomous vehicles and robotic mapping applications. However, platform motion during acquisition may introduce geometric distortion within the resulting point clouds [25].

To evaluate the quality and consistency of point clouds, distance-based comparison methods are commonly used. One common approach is C2C comparison, where distances are computed between corresponding or nearest points in two point clouds [23]. This provides a direct measure of geometric deviation between point clouds. Registration techniques such as ICP are commonly applied beforehand to spatially align point clouds by minimising the distance between corresponding regions. More advanced methods, such as M3C2, additionally account for local surface geometry during distance estimation [24].

3.3 Active Gimbal Systems for UGVs

Active gimbal systems are mechanical stabilisation systems designed to maintain a desired sensor orientation during platform motion, typically using one or more actively controlled rotational axes [11]. By mechanically decoupling the sensor payload from external disturbances, gimbal systems reduce unwanted sensor motion and help maintain stable sensor alignment during dynamic operation. Such stabilisation systems are commonly used for camera and sensor payloads in applications where platform motion may otherwise degrade measurement quality. In mobile robotic systems, terrain-induced motion may propagate directly to rigidly mounted sensors, reducing measurement consistency and perception performance [5], [12]. Active stabilisation, therefore, becomes particularly relevant for UGVs operating in uneven terrain environments.

Software-based compensation methods, including Kalman-filter-based approaches, may partially reduce motion-induced distortion through filtering and post-processing. However, such approaches remain limited by sensor latency, sampling constraints, and information loss occurring during measurement acquisition [26], [27]. Mechanical stabilisation through active gimbals instead reduces physical sensor movement prior to measurement acquisition [11], potentially improving spatial consistency and temporal coherence in acquired sensor data, which is important for perception and localisation performance [27].

Active gimbal stabilisation has therefore become an important design consideration for robotic systems relying on high-quality vision- and LiDAR-based perception under dynamic operating conditions [11]. For UGVs, the integration of such systems presents unique challenges due to increased vibration levels. The lightweight SLaRC UGV additionally imposes constraints on platform size, weight, and mechanical integration. Nevertheless, active stabilisation remains a promising approach for improving perception robustness during uneven terrain traversal.

3.4 Control Strategies

Feedback control constitutes a fundamental component of robotic systems dealing with dynamic environments [28], [29]. For actively stabilised sensor platforms on UGVs, the primary control objective is to maintain a desired sensor orientation. Such stabilisation requires controllers that are robust, computationally efficient, and suitable for real-time applications.

Among the available control approaches, PID controllers remain one of the most widely adopted control strategies in engineering applications due to their simple structure, ease of implementation, and robust performance across a broad range of operating conditions. In many robotic and mechatronic applications, the low computational complexity and predictable behaviour of PID-based controllers make them preferable to more advanced methods, particularly when real-time constraints and limited onboard computational resources are present [28], [29]. Alternative control strategies, including robust, optimal, adaptive, and non-linear control methods, have also been proposed for stabilisation problems involving modelling uncertainty and external disturbances [29], [30], [31]. Although such approaches may provide improved robustness or disturbance rejection under certain conditions, they generally require more accurate system modelling, increased tuning complexity, or higher computational requirements compared to classical PID control. Consequently, classical PID control remains a practical and widely adopted approach for active stabilisation systems operating under real-time constraints.

3.5 Attitude Estimation

Accurate attitude estimation is fundamental for UGVs operating in challenging terrain. Attitude information, expressed as roll, pitch, and yaw, directly affects vehicle stability, control, and orientation of onboard sensors [32], [33]. The observability of orientation differs between the rotational axes. Roll and pitch can be directly referenced to the gravity vector through accelerometer measurements, providing an absolute physical reference [26], [27].

Yaw differs fundamentally from roll and pitch in that it lacks a direct inertial reference when relying solely on accelerometers and gyroscopes. In stand-alone inertial systems, yaw is typically obtained through integration of angular rate measurements, which leads to drift over time [26], [34]. In the SLaRC UGV platform presented in Section 5.3.1, yaw heading is obtained from Global Positioning System (GPS)-based heading estimation, providing an external absolute reference[32], [33].

Attitude estimation for mobile robotics is most commonly achieved using IMUs, which combines tri-axial gyroscopes and accelerometers to measure angular velocity and linear acceleration. Gyroscopes provide high-rate angular velocity measurements that enable smooth short-term orientation tracking; however, their integration over time leads to drift due to bias and noise accumulation. Accelerometers provide a gravity-referenced correction mechanism but are highly sensitive to linear accelerations induced by braking, suspension motion, and terrain impacts [26], [32], [35]. As a result, neither sensing modality is individually sufficient, and attitude estimation relies on fusion techniques that exploit their complementary characteristics.

In the context of UGVs, inertial sensing is subject to additional challenges due to the roll and pitch variations caused by the terrain. These dynamics degrade the accelerometer measurements and increase estimation uncertainty. As a result, attitude estimation approaches for UGVs must remain stable under sustained dynamic movement, rather than relying on quasi-static assumptions commonly employed in simpler filtering methods [36]. To address these challenges, a wide range of filtering approaches have been developed, including complementary filters, Kalman filters, and non-linear observers. These methods aim to balance short-term responsiveness with long-term stability while maintaining robustness to noise and dynamic disturbances [33], [35].

4. Related Work

Related work on active sensor stabilisation spans multiple research areas, including mechanical gimbal systems, control strategies for disturbance rejection, and IMU attitude estimation. This section reviews prior work relevant to the design of actively stabilised sensor platforms for mobile robotic applications, with a particular focus on two-axis gimbal systems and IMU-based orientation estimation. Together, the reviewed works provide the foundation for the methodology adopted in this thesis.

4.1 Stabilised Platforms and Gimbal Systems

Actively stabilised platforms and gimbal systems constitute a common mechanical approach for mitigating sensor motion induced by disturbances in mobile robotic platforms [11], [37], [38], [39]. By actively compensating for rotational motion, such systems aim to preserve a desired sensor orientation relative to gravity. This subsection reviews prior work on stabilised platforms with particular emphasis on low-cost IMU-based implementations and inertially stabilised gimbals, as these approaches are most relevant to the design choices and system constraints considered in this thesis.

4.1.1 Low-cost IMU-based Stabilised Platforms

Active stabilised platforms have been widely studied in applications where sensors or payloads are subjected to external motion disturbances. Tripathi and Gupta [40] present a self-stabilising platform intended for use in moving vehicles in the medical field, such as boats, helicopters, and cars. Although the application differs from UGV operation, their work demonstrates the fundamental principle of combining IMU-based feedback together with motorised actuation to maintain platform stability under external disturbances. Their system relies on a single IMU (MPU6050) [41] providing three-axis motion measurements. The combination of an IMU with servo or DC motor actuation to counteract disturbances illustrates a common low-cost approach to active platform stabilisation.

4.1.2 Inertially Stabilised Multi-DOF Gimbal Systems

Inertially stabilised gimbal systems decouple sensor orientation from platform-induced motion by actively compensating rotational disturbances using inertial sensing and closed-loop control. 2-DOF gimbals, typically compensating for roll and pitch, are commonly employed in ground-based platforms where tilt disturbances dominate sensor misalignment. Popelka [42] investigates a hardware-implemented prototype 2-DOF stabilising platform using an IMU-based sensing approach and highlights the importance of sensor placement. Mounting the IMU on the stabilised platform improves observability of residual angular motion, but increases sensitivity to noise and vibration. Conversely, placing the sensor on the base reduces noise at the cost of reduced estimation accuracy. This trade-off is particularly relevant for mobile robotic systems, where both measurement quality and robustness to vibration are important factors to consider. In Popelka's proposed stabilisation architecture, the IMU is mounted on the stabilised platform to prioritise direct measurement of residual roll and pitch disturbances affecting the LiDAR payload.

PID-based stabilisation has been used in two-axis gimbal systems due to its simplicity and reliability in real-world systems [28], [29], [37]. Ahmad et al. [37] present a detailed dynamic model and PID controller design for a 2-DOF inertially stabilised gimbal, demonstrating effective disturbance rejection under varying base angular motion. However, their approach relies on simplified modelling assumptions and trial-and-error tuning, with validation primarily limited to simulation rather than experimental hardware evaluation. Similarly, Ghaeminezhad et al. [43] employ a conventional PID controller as the baseline stabilisation method for a similar two-axis gimbal platform subjected to base-induced disturbances, demonstrating that PID control provides sufficient tracking and disturbance rejection under nominal conditions in a simulation environment. While advanced control extensions have been proposed, these studies reinforce PID control as a practical and commonly adopted solution for roll-pitch stabilisation systems.

Three Degrees of Freedom (3-DOF) gimbal systems extend stabilisation to include yaw, enabling full orientation decoupling from platform motion. Such systems are prevalent in aerial and maritime robotics, where rapid heading changes and external disturbances significantly affect sensor alignment [38], [44], [45]. Liu et al. developed and experimentally evaluated a three-axis inertially stabilised gimbal for a Unmanned Aerial Vehicle (UAV)-mounted camera, demonstrating stable line-of-sight tracking under dynamic flight conditions [39], demonstrating the effectiveness of 3-DOF inertial stabilisation in aerial applications. Similarly, Xin et al. evaluated a three-axis gimbal for a surface vehicle operating in heavy sea conditions, showing that active mechanical stabilisation substantially reduces platform-induced motion compared to rigid mounting [38].

Although operational context differs, the underlying stabilisation principles remain consistent across domains. Multi-axis inertial stabilisation aims to mitigate platform-induced rotational disturbances through active compensation in the relevant DOF. However, the necessity of yaw stabilisation depends strongly on application and domain requirements. In aerial and maritime systems, heading disturbances directly influence sensor line-of-sight and global orientation. In ground-based robotic systems, roll and pitch motion dominantly affect gravity-referenced sensor orientation and inertial alignment, making these axes particularly relevant for stabilised perception systems [46].

4.1.3 Robust Solutions for Harsher Applications

More application-specific work is presented by Iqbal et al. [47], who describe a 2-DOF stabilisation platform designed for marine environments, such as antenna stabilisation on ocean-going vessels. Additionally, they suggest using H_∞ based methods to design a robust controller. An H_∞ feedback controller minimizes the worst-case effect of disturbances, noise, and model uncertainty on a system. It is designed to maintain stability and performance even when the real system differs from the mathematical model. Through simulated and real-world experiments, they demonstrated that an H_∞ based controller was suitable to maintain robustness, as well as maintain good performance for their application. Their platform consisted of a 2-DOF parallel manipulator designed to reject rotational disturbances and maintain a level top plate relative to the horizon. Two variable-length actuators connected to the base and top plates, while onboard sensors provided angular measurements for closed-loop control. While their mechanical design differs from conventional serial gimbal systems, their work highlights the importance of robust control strategies when operating in highly disturbed environments.

Görmüş et al. [45] investigate passive vibration isolation for gimbal-mounted surveillance systems on mini UAVs. Their work demonstrates how mechanical isolation can reduce vibration transmission from the vehicle platform to stabilised sensor payloads, thereby improving perception quality. Although the operational context differs from ground-based UGVs, the study highlights the importance of vibration mitigation and mechanical stabilisation for maintaining sensor measurement quality under dynamic operating conditions.

4.2 Control Strategies

The control architecture of stabilised platforms typically consists of nested control loops, where an inner stabilisation loop is responsible for disturbance rejection and an outer loop handles tracking or higher-level objectives. Baskin and Leblebcioglu [48] emphasize that effective disturbance rejection is critical for minimizing the impact of vibrations and external disturbances, particularly when low-cost Micro Electro Mechanical Systems (MEMS) sensors are used. Measurement delays, sensor noise, and limited bandwidth are identified as key performance-limiting factors in such systems. Filtering techniques, including notch filters, may additionally be applied to attenuate structural resonances and vibration-induced disturbances. Baskin and Leblebcioglu additionally investigate H_∞-based disturbance observers for improved disturbance estimation and rejection under sensor noise and modelling uncertainty.

Chen et al. [49] propose a controller design for two-axis gimbal platforms based on linear matrix inequality (LMI) optimisation, enabling finite-time stabilisation while explicitly accounting for cross-coupling between the pitch and yaw axes. Their work models the gimbal system as a linear parameter-varying (LPV) system and demonstrates the effectiveness of LMI-based control

in achieving fast convergence and robustness to disturbances. The authors also describe a common gimbal control structure consisting of an outer tracking loop and an inner line-of-sight (LOS) stabilisation loop, where the inner loop is primarily responsible for disturbance rejection.

Compared to PID-based approaches, robust control methods such as H_∞ optimisation have been shown to better handle multivariable interactions and unmodeled dynamics in two-axis gimbal systems [28], [29], [47], [48]. However, these methods often require accurate system modelling and increased computational complexity. Alternative non-linear approaches, such as Sliding Mode Control (SMC), have also been applied to LOS stabilisation but may introduce high-frequency switching behaviour and increased noise sensitivity, which can be problematic in practical implementations.

4.3 LiDAR-Based Perception and Stabilisation

Existing work has investigated the integration of LiDAR sensors with active gimbal systems to improve perception robustness. These studies investigate both mechanical stabilisation approaches and control strategies together with their impact on mapping and SLAM performance. The following subsections review representative implementations and comparative evaluations.

4.3.1 Gimbal-mounted LiDAR for SLAM

Shigenfuji et al. [50] investigated a three-axis gimbal-mounted LiDAR system for marker-based LiDAR-SLAM. Their system actively rotated the LiDAR using the gimbal to increase environmental coverage, with point clouds registered according to the measured gimbal orientation. The authors compared the gimbal-mounted configuration against a rigid chassis-mounted configuration combined with an Attitude and Heading Reference System (AHRS)-based orientation estimation system. Both approaches achieved point cloud acquisition accuracies of approximately 0.04 m under the evaluated conditions. The study therefore demonstrates that both mechanical stabilisation and inertial attitude compensation may improve perception performance in dynamic environments. However, different LiDAR sensors were used in the two configurations, limiting the validity of the comparison between rigid and actively stabilised mounting approaches.

4.3.2 Comparison of gimbal-mounted and rigidly mounted LiDAR

Trybala et al. [20] developed a mobile mapping system using a Livox-based LiDAR mounted on an actively stabilised gimbal platform. The stabilisation system was intended to reduce motion-induced disturbances and improve point cloud consistency during mobile operation in environments such as forests and underground tunnels. An experimental comparison between gimbal-mounted and rigidly mounted configurations showed that the stabilised configuration measured gyroscope noise density approximately 86 % and accelerometer noise density by $31 - 50$ % depending on axis. The authors further demonstrated centimetre-level mapping accuracy in challenging environments using the stabilised configuration. These results support the premise that mechanical stabilisation can improve inertial measurement stability and reduce motion-induced degradation in LiDAR-based perception systems.

4.3.3 Modelling and control aspects for LiDAR gimbals

Huang et al. [7] investigated dynamic modelling and control of a two-axis (yaw and pitch) stabilised gimbal system. The authors proposed a Kane-based modelling framework together with an optimised PID-based controller for improved tracking and disturbance rejection performance. Their work demonstrates the importance of accurate dynamic modelling when analysing multi-axis stabilised platforms, particularly in the presence of non-linear motion and external disturbances. The proposed controller was evaluated in simulation and compared against conventional PID-based approaches. The optimised controller demonstrated improved tracking performance and reduced overshoot during dynamic motion conditions. Although the study focuses on yaw-pitch stabilisation and advanced controller optimisation, the work supports the relevance of closed-loop disturbance rejection and dynamic modelling for actively stabilised sensor platforms.

4.3.4 Point Cloud Accuracy and Evaluation

Point cloud processing plays a pivotal role in evaluating LiDAR-based perception systems. A wide range of methods have been developed to assess geometric consistency and detect changes between point cloud datasets. One of the most widely used approaches is the ICP algorithm, which aligns two point clouds by minimising the distance between corresponding points. This method is commonly combined with C2C distance computation to quantify geometric differences between datasets [23].

While C2C method provides a straightforward measure of deviation, they are sensitive to noise, point density variations, and surface roughness. To address these limitations, more advanced approaches such as the M3C2 method have been proposed. M3C2 computes distances along surface normals and incorporates local surface characteristics, allowing for more robust and accurate comparison in complex 3D environments [24].

However, despite these post-processing advancements, the accuracy of point cloud comparison remains strongly influenced by registration errors and sensor motion during data acquisition. Lague et al. demonstrate that misalignment between datasets can dominate the overall error, often exceeding the intrinsic measurement noise of the LiDAR itself [24]. A key limitation is that post-processing techniques improve alignment but cannot fully correct errors originating during data acquisition, demonstrating the importance of accurate sensor stabilisation for maintaining point cloud geometric consistency.

4.4 Experimental Evaluation of Stabilised Sensor Platforms

Evaluation methodologies for actively stabilised sensor platforms vary depending on the application domain, but several common experimental patterns can be identified across aerial, maritime, and ground-based systems. In aerial applications, stabilised gimbals are commonly evaluated under vibration and attitude disturbances generated during UAV flight. Görmüş et al. [45] experimentally assessed vibration damping performance of an electro-optical/infrared (EO/IR) gimbal mounted on a UAV platform by measuring vibration transmission and quantifying the effectiveness of the mechanical isolation system. Such evaluations quantify attenuation of platform-induced excitation and analyse the impact on imaging stability. In contrast, in maritime environments, stabilisation systems are subjected to roll, pitch, and yaw disturbances induced by wave motion. Xin et al. [38] experimentally validated a three-axis gimbal mounted on a surface vehicle by subjecting the system to controlled motion disturbances and analysing line-of-sight tracking performance. Their experimental study included time-domain tracking responses and disturbance rejection analysis, demonstrating improved orientation stability compared to baseline control approaches. Liu et al. [39] implemented and experimentally validated a three-axis inertially stabilised gimbal under dynamic motion conditions. Their evaluation involved analysing tracking error and time-domain response during dynamic platform motion. For ground-based robotics, Lee et al. [11] conducted a comparative experimental study of two-axis gimbal controllers using motion tracking and target tracking tasks to assess disturbance suppression and tracking precision between a robust double active controller and an integral SMC. In addition, Ahmad et al. [37] simulated a PID-based two-axis gimbal stabilisation system using step-response and disturbance tests to evaluate pointing stability and tracking accuracy.

Across these domains, experimental methodologies consistently involve:

1. Controlled disturbances excitation
2. Closed-loop tracking evaluation
3. Time-domain response analysis
4. Quantitative performance comparison

5. Method

The following subsections describe how the project methods were utilised in each phase.

5.1 Research Methodology

This thesis adopts a systems development research methodology, in which scientific knowledge is generated through the design and evaluation of a technical system [51]. Systems development has been identified as a valid research methodology within applied and engineering-oriented research, particularly when the research objective is to investigate the feasibility, performance, and implications of a proposed technical solution. Within this approach, the development of a technical system is used as a means to investigate and evaluate a research problem through iterative design and quantitative analysis. This methodology directly supports the formulated research questions presented in Section 2. by enabling quantitative evaluation of roll and pitch attenuation performance (RQ1), derivation of stabilisation accuracy requirements (RQ2), and quantitative comparison between rigid and actively stabilised configurations (RQ3).

In this thesis, the methodology was applied through the design and modelling of a proposed actively stabilised sensor platform, combined with simulation-based evaluation using experimentally recorded terrain traversal attitude data acquired from the SLaRC UGV. The methodology further enabled quantitative analysis of disturbance rejection performance, residual stabilisation error, and resulting LiDAR geometric consistency under representative terrain-induced disturbances.

5.2 Project Management

To support project planning and identify potential technical and scheduling risks, a risk analysis was conducted during the initial phase of the project. The complete risk analysis is presented in Appendix B. Following the risk analysis, a WBS was developed to decompose the project into manageable tasks and development phases, including literature study, system design, modelling, evaluation, and documentation. The WBS is shown in Appendix A. Based on the WBS, a Gantt chart was developed to support scheduling and time management across the 20-week project duration. The Gantt chart is presented in Appendix C.

5.3 System Architecture and Platform Context

The adopted methodology in this project, together with the tools and methods for each phase, is illustrated in Figure 2.

5.3.1 The SLaRC UGV Platform

The SLaRC UGV platform, developed in a prior study, serves as the experimental baseline platform used throughout this thesis. The platform consists of an onboard ASUS NUC 13 computer, a dedicated power distribution system, and four in-wheel motors providing skid-steer mobility with integrated suspension. The software architecture was developed in Robot Operating System 2 (ROS2), supporting sensor data acquisition, controller development, and simulation-based evaluation. The overall system architecture and communication pathways are illustrated in Figure 3.

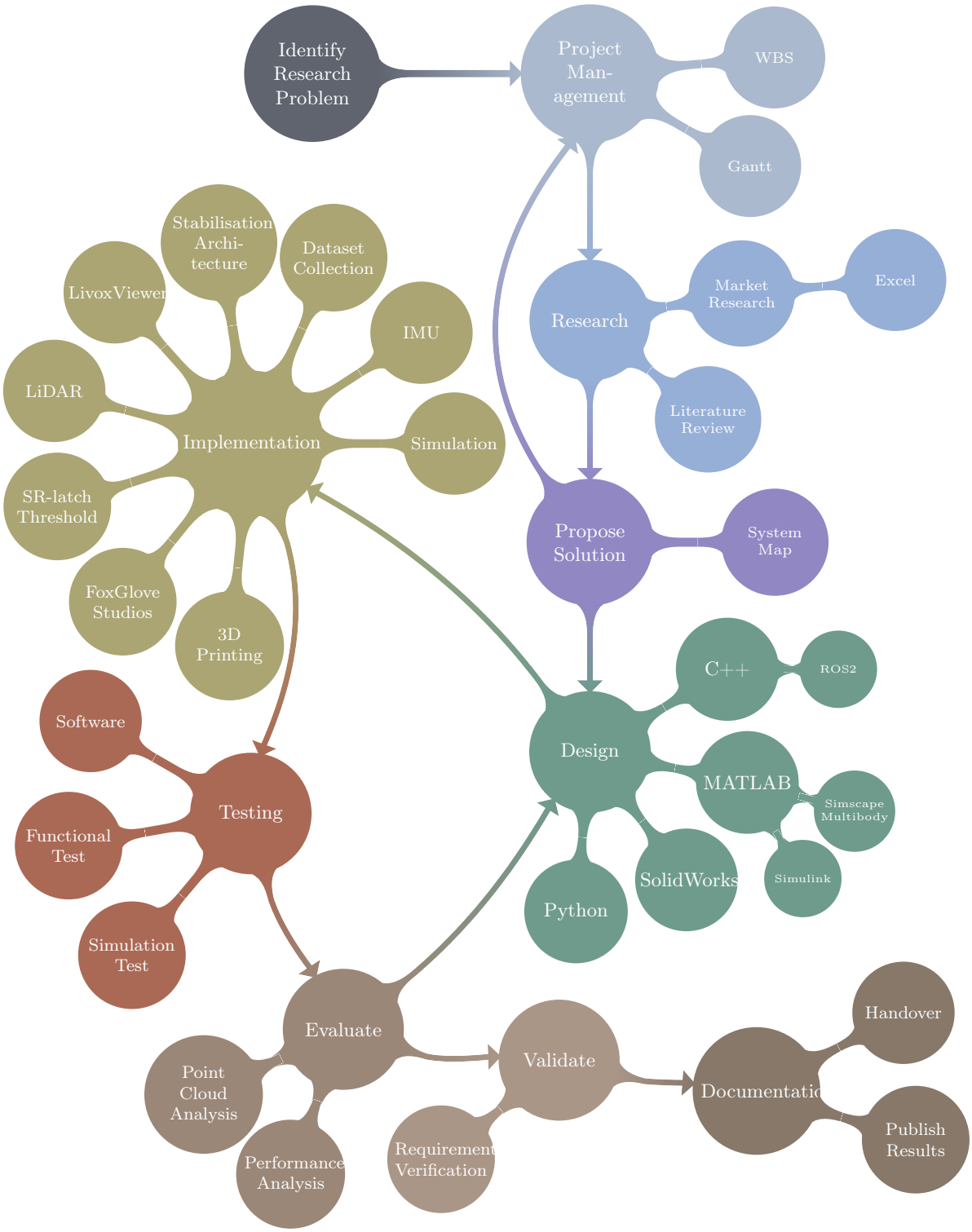


Figure 2: The figure illustrates an overview of the project workflow. Each color-code is associated with a specific method and its corresponding tools and techniques. The working process began with stating the research problem. The following phase included a literature review and a market research, resulting in a proposed solution. The next step was an iterative phase were implementation, testing and evaluation would either lead to re-design or validation against the requirements. Lastly the process and results where documented both as a technical report but also for handover purposes.

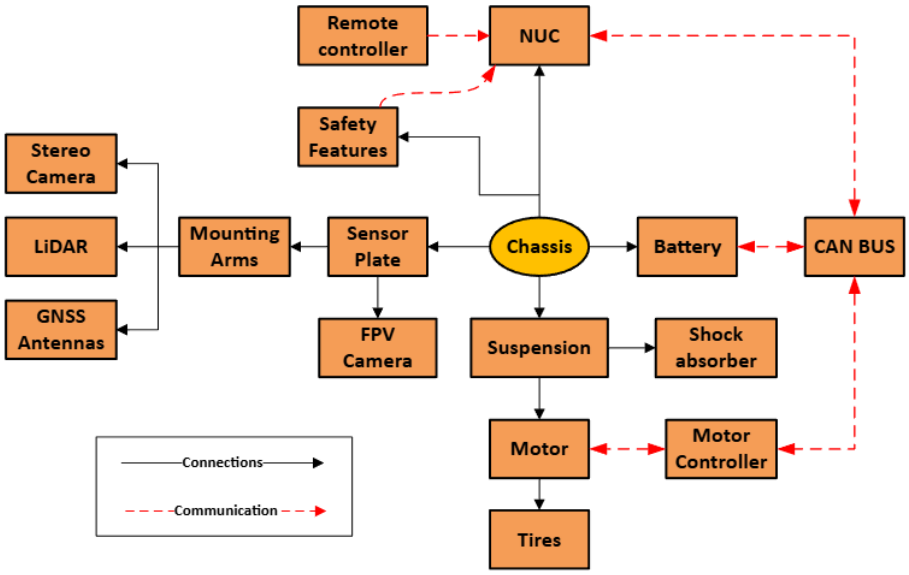


Figure 3: System-level overview of the SLaRC UGV platform, showing the onboard computing unit, power distribution, motor control, sensor interfaces, and communication pathways that form the baseline system used in this thesis.

Within the scope of this thesis, the SLaRC UGV platform is treated as a fixed baseline platform. Consequently, modifications to the vehicle chassis, suspension dynamics, drivetrain control, autonomous navigation stack, and LiDAR-based SLAM algorithms are considered outside the scope of this work. Furthermore, the proposed stabilisation platform is not inherently dependent on the SLaRC UGV, although the platform was selected due to its availability and existing sensor integration infrastructure. The baseline UGV configuration and component placement are shown in Figure 4. Instead, the focus of this work was limited to the design, modelling, and simulation-based evaluation of an actively stabilised sensor platform mounted on top of the UGV, with the goal of improving LiDAR measurement stability during terrain traversal. The SLaRC UGV utilised a

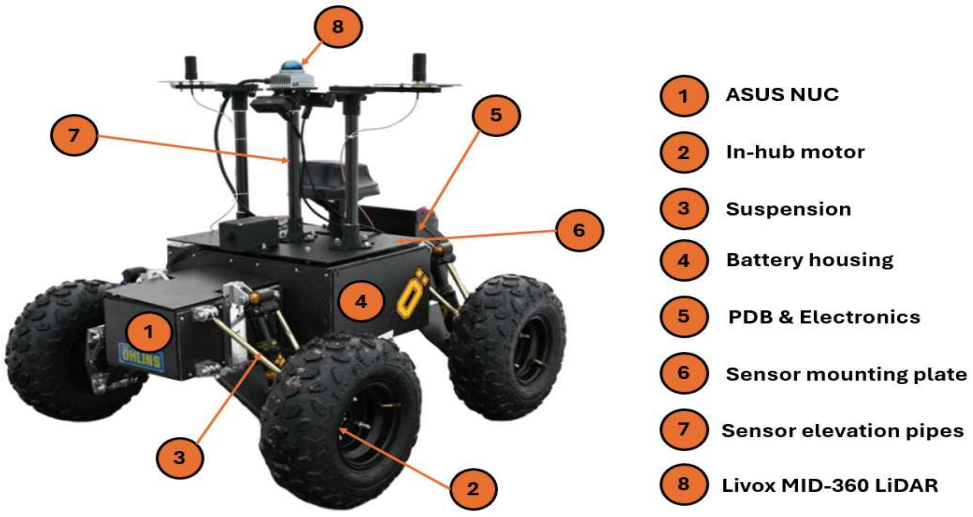


Figure 4: Physical overview of the SLaRC UGV platform, highlighting the chassis, suspension system, drivetrain, and sensor mounting locations.

rigid roof-mounted sensor assembly designed to support modular sensor integration while maintaining structural rigidity during operation. The primary sensor of interest is the Livox MID-360 LiDAR [52], mounted directly to the rigid sensor platform. The LiDAR configuration was treated as a fixed baseline sensor arrangement throughout the evaluation framework.

Since the sensors were rigidly mounted to the vehicle chassis, their orientation directly followed the motion of the UGV. During uneven terrain traversal, the UGV experiences significant roll and pitch motion caused by obstacles, slopes, and surface irregularities. Although the suspension system attenuates high-frequency vibrations, it does not decouple the sensor frame from the low-frequency chassis attitude variations. Consequently, terrain-induced chassis motion may introduce geometric distortion into the acquired LiDAR measurements during traversal. This limitation motivated the development of the proposed actively stabilised sensor platform to mitigate these attitude variations. The rigid baseline sensor configuration is illustrated in Figure 5.



Figure 5: Rigid sensor mounting solution on the SLaRC UGV, where sensors are fixed directly to the chassis without active orientation compensation.

5.4 Design Decisions

Previous work presented in Section 4. demonstrated the feasibility of low-cost IMU-based active gimbal stabilisation platforms using servo actuation and complementary filtering. However, these systems are commonly evaluated in stand-alone configurations without representative terrain-induced disturbance input or quantitative evaluation of resulting perception accuracy. Performance assessment is also often limited to qualitative stabilisation behaviour rather than quantitative analysis of residual stabilisation error and resulting LiDAR geometric distortion [42]. To evaluate stabilisation performance in relation to perception quality, the dominant sources of geometric distortion within ground-based LiDAR perception must first be identified. For ground-based mobile robotic platforms, yaw represents the rotation about the gravity vector and therefore does not affect sensor gravity alignment. In contrast, roll and pitch directly tilt the sensor frame relative to gravity, introducing range-dependent geometric distortion within LiDAR measurements during uneven terrain traversal.

The geometric effect of sensor tilt on range-based perception can be approximated by considering a LiDAR observing a target at horizontal range R. A roll or pitch orientation error θ introduces a vertical deviation approximately given by the following equation:

$$\Delta h \approx R \tan(\theta), \tag{1}$$

where the resulting vertical deviation increases proportionally with sensing range. Here, θ denotes the effective angular misalignment of the LiDAR sensor relative to gravity. Consequently, even small residual attitude errors may produce significant geometric deviation at longer ranges. In contrast, yaw rotation occurs about the gravity vector and does not directly affect the vertical projection of the sensed environment. Motion-induced geometric distortion in scanning LiDAR systems has been documented in the literature [9], [10].

For ground-based UGVs, yaw motion is primarily associated with steering and navigation rather than terrain-induced attitude disturbance. Three-axis stabilisation systems are therefore more commonly utilised in aerial and maritime applications, where continuous heading and elevation

disturbances strongly influence sensor orientation. In contrast, terrain-induced disturbances on ground-based platforms predominantly manifest as roll and pitch motion. Introducing an additional yaw stabilisation axis would further increase mechanical inertia, actuator coupling, and control complexity while providing limited improvement in gravity-aligned LiDAR perception performance with respect to vertical attitude error [7], [38]. Consequently, this thesis focused on active roll and pitch stabilisation to address the dominant sources of LiDAR geometric distortion in uneven terrain environments.

5.4.1 Sensing Architecture and Requirements

Reliable attitude estimation is required to maintain stable LiDAR orientation during terrain traversal and minimise range-dependent geometric distortion caused by residual platform tilt. The sensing requirements were derived from the allowable geometric deviation introduced by residual roll and pitch error, as established in Section 5.4. A vertical constraint, for example, of 5 cm at a sensing range of 30 m corresponds to a maximum permissible tilt error of approximately 0.1° based on equation 1. For combined roll and pitch motion, the allowable residual attitude error was expressed as the root-sum-square magnitude of the two-axis RMS tilt error.

$$\sqrt{\phi_{\mathrm{RMS}}^2 + \theta_{\mathrm{RMS}}^2} \leq 0.1^\circ, \tag{2}$$

ensuring that the resulting geometric deviation remained within the derived LiDAR accuracy constraint.

Experimental studies of UGV suspension dynamics, including the work by Rubiec et al. [21], indicate that terrain-induced roll and pitch disturbances are predominantly concentrated at low frequencies during uneven terrain traversal. These disturbances are typically observed below approximately 10, Hz, defining the primary disturbance bandwidth relevant to stabilisation performance for the SLaRC UGV's intended use. To achieve effective disturbance rejection, the stabilisation controller bandwidth was therefore required to exceed the dominant terrain-induced disturbance frequencies while maintaining sufficient closed-loop stability margins. Based on these observations, a target closed-loop stabilisation bandwidth of approximately 10–15, Hz was considered sufficient for the representative disturbances. To preserve phase margin and minimise discretisation-induced delay within the digital feedback loop, the IMU orientation update rate was required to significantly exceed the closed-loop stabilisation bandwidth. In sampled-data control systems, sampling frequencies are commonly selected to be several times higher than the closed-loop bandwidth to reduce phase degradation introduced by discretisation [53]. Consequently, an IMU orientation output rate above approximately 100, Hz was considered sufficient for the proposed stabilisation framework.

Low-cost MEMS IMUs typically provide raw accelerometer and gyroscope measurements requiring external sensor fusion and filtering to obtain drift-compensated orientation estimates. Development of a custom inertial navigation and sensor fusion framework was considered outside the scope of this thesis, which primarily focused on mechanical stabilisation for improved geometric consistency performance for a LiDAR perception. Accordingly, industrial fused Inertial Navigation System (INS)-class IMUs with embedded sensor fusion were preferred for the proposed stabilisation framework. Such systems provide internally filtered attitude estimates with validated sensor fusion architectures, reducing the need for custom filtering implementation within the stabilisation controller. Compared to lower-cost raw MEMS alternatives, industrial fused IMUs generally provide improved bias stability, noise characteristics, and validated orientation estimation performance [54]. Given the emphasis on stabilisation accuracy and representative perception evaluation rather than low-cost hardware experimentation, fused INS-class IMUs satisfying the derived update-rate and attitude accuracy requirements were therefore considered appropriate for the proposed system design.

5.5 Performance Metrics and Evaluation Criteria

The selected performance metrics were chosen to evaluate both closed-loop stabilisation behaviour and the resulting impact on LiDAR measurement quality. Prior work on inertially stabilised gimbal systems commonly evaluates stabilisation performance in terms of disturbance rejection, pointing

accuracy, tracking stability, and vibration attenuation [7], [11], [38], [48]. In addition, previous research has shown that platform motion and sensor actuation influence the geometric quality and consistency of reconstructed LiDAR point clouds [18].

To enable an objective evaluation against the rigidly mounted baseline, quantitative performance metrics were defined prior to the simulation-based analysis.

Angular RMS Error: The angular RMS error was defined as the root-mean-square value of the stabilised roll or pitch error evaluated over N sampled measurements. RMS error was selected because stabilisation under terrain-induced motion constitutes a continuous disturbance rejection problem, where sustained residual motion is more representative than isolated peak deviations. Orientation tracking accuracy was evaluated using RMS angular error:

$$\theta_{\mathrm{RMS}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} e_i^2} \tag{3}$$

where e_i denotes the instantaneous orientation error between desired gravity alignment and measured platform orientation. This metric quantifies sustained stabilisation accuracy under dynamic disturbance.

Stability Detection: Platform stability was evaluated using the binary stability signal S_{stable}, derived from the combined roll and pitch stability conditions described in Section 7.1.4. Stable operation was quantified using the percentage of time for which the platform remained within the defined stability thresholds, together with the duration of continuous stable intervals.

DRR: Disturbance rejection capability was quantified as

$$\mathrm{DRR} = \frac{\theta_{\mathrm{RMS,base}}}{\theta_{\mathrm{RMS,platform}}} \tag{4}$$

Higher DRR values indicate improved attenuation of terrain-induced motion. Both RMS values were computed over the same time window and under identical disturbance conditions to ensure that the DRR reflects true attenuation performance rather than differences in excitation or measurement duration.

LiDAR Vertical Deviation: To translate angular stabilisation performance into spatial geometric error relevant to LiDAR perception, the vertical deviation was calculated using Equation 1 where R represents the sensing range and θ denotes the residual angular error.

6. Ethical and Societal Considerations

The stabilisation platform developed in this thesis cannot be viewed in isolation from its system context. The SLaRC UGV is directly funded by the Swedish defense sector, and the sensor platform designed and implemented in this thesis is mounted directly on top of this UGV and intended to enable high-accuracy LiDAR perception in challenging terrain. Even though the immediate technical focus is limited to simulation, control for sensor stabilisation, inertial sensing, and mechanical design, the resulting capability can in practice be integrated into defense-related applications as well as civilian use cases. This close connection to a defense-affiliated platform, combined with the dual-use nature of high-fidelity perception, motivates a broader discussion of ethical, legal, societal, and environmental aspects beyond the purely engineering-oriented contributions of this work.

6.1 Defence Context and Ethical Positioning

This thesis is intended for educational and research purposes; however, it is written for SLaRC, which is sponsored by the Swedish Defence Materiel Administration (FMV). FMV is responsible for purchasing the material needed by the Swedish Armed Forces, which includes combat equipment and weapon systems. Furthermore, the UGV used in this project is constructed within the framework of the ARES Makeathon, which is a competition held by the European Defense Agency (EDA) to promote the development of UGVs used for military purposes. The technology developed and evaluated in this thesis can be used for both military and civilian domains. Military use cases include reconnaissance and operational assistance, while civilian applications include agricultural maintenance, environmental monitoring, and infrastructure inspection. This means that even if the intended usage of the results in this project is for educational and research purposes, there is a possibility that the findings of this thesis will be used for military purposes. In addressing this dual-use potential, the work is positioned as a generic stabilisation and perception enhancement framework without application-specific optimisation for weaponisation or offensive capabilities.

6.2 Economic and Labour Implications

The increasing deployment of autonomous systems such as UGVs introduces significant economic and labour-related implications, particularly regarding labour displacement and changing workforce requirements [55], [56]. Automation is expected to affect routine manual and transportation-related occupations most strongly, where autonomous systems may replace tasks previously requiring human intervention. Research indicates that increased industrial robot adoption correlates with measurable reductions in employment within certain sectors [55]. In addition, automation may contribute to skill-biased technological change, where demand increasingly shifts toward highly skilled labour while middle-skilled occupation decline [57].

At the same time, autonomous technologies can improve productivity, reduce operational costs, and create new technical and engineering roles associated with development, deployment, and maintenance of robotic systems. The societal impact of UGV deployment therefore depends on how the economic benefits of automation are balanced against its labour market consequences [55], [57].

6.3 Legal and Regulatory Perspectives

The deployment of UGVs in civilian and commercial environments raises several legal and regulatory considerations related to system safety, accountability, and data protection. From a liability perspective, autonomous and semi-autonomous robotic systems introduce difficult questions regarding responsibility in the event of malfunction or accidents, particularly concerning the roles of operators, manufacturers, and system integrators [58], [59]. Emerging regulatory frameworks, including the European Union Artificial Intelligence Act, increasingly emphasise risk management, transparency, and clear allocation of accountability for high-risk Artificial Intelligence (AI)-enabled systems.

Data protection and privacy constitute particularly relevant concerns for sensor-equipped UGVs, since systems relying on cameras, LiDAR, and environmental sensing may unintentionally capture personal data or sensitive information during operation [57]. Such systems must therefore comply

with applicable data protection regulations governing the collection, storage, and processing of sensor data in public environments.

6.4 Environmental and Sustainability Considerations

UGVs also raise broader questions about sectoral sustainability and environmental impact. Life-cycle assessments indicate that in some domains robotic systems can reduce environmental pressures compared to conventional techniques: for example, autonomous weeding platforms in agriculture have been shown to cut both greenhouse-gas emissions and pesticide use relative to traditional chemical weed control [60]. At the same time, these gains are counterbalanced by the material intensity of robotic hardware: batteries, power electronics, semiconductors, and rare-earth-based components entail non-trivial ecological burdens during extraction, manufacturing, and end-of-life treatment, especially when robust reuse and recycling pathways are absent [61].

Similar trade-offs appear in industrial and logistics settings. Automation can enable more efficient production planning and process control, reducing energy waste and improving utilisation of raw materials [62]. However, these efficiency improvements may also trigger rebound effects, where lower unit costs and higher throughput stimulate additional demand, thereby increasing overall energy use and material consumption and potentially undermining climate and resource-efficiency targets. Almusharaff argues that robotics and automation should therefore be explicitly embedded in circular-economy strategies, emphasising design principles such as modularity, ease of repair, extended product lifetimes, and ethically sourced materials [63]. Aligning UGV, and by proxy sensor stabilisation platforms, development with such principles, and with EU-level sustainability initiatives, is essential to ensure that efficiency gains are not outweighed by ecological costs and that the technology is perceived as both environmentally and socially responsible. In the longer term, the trajectory of UGV deployment will be shaped not only by technical progress but also by regulatory frameworks and public expectations, which together will determine whether these systems are accepted as trustworthy and sustainable technologies [64].

7. Implementation

This section describes the implementation of the actively stabilised sensor platform based on the design requirements established in Section 5.4. The implementation includes the simulation environment, IMU-based attitude estimation, closed-loop PID control architecture, stability detection framework, and the LiDAR-based point cloud evaluation pipeline used for geometric evaluation. The overall control objective was to maintain a gravity-aligned LiDAR frame independent of chassis orientation during traversal in uneven terrain on the SLaRC UGV.

The evaluation framework proposed in this thesis employs controlled terrain-induced disturbances, closed-loop roll-pitch tracking analysis, RMS and disturbance rejection metrics, and comparative assessment against a rigidly mounted baseline configuration. Such methodologies provide measurable indicators of stabilisation performance and enable comparative assessment of control strategies under realistic operating conditions.

7.1 Simulation of PID-based Stabilisation Controller

The proposed stabilisation strategy was evaluated in simulation to tune controller parameters and assess closed-loop behaviour under representative terrain disturbances. The simulation model was limited to roll and pitch motion, corresponding to the dominant contributors to vertical LiDAR pointing error. Yaw stabilisation was excluded since the research questions focused specifically on the dominant contributors to vertical pointing LiDAR error, roll, and pitch. A classical PID controller per axis was selected on this basis, following the architecture commonly adopted for 2-DOF inertially stabilised gimbal systems [28], [29], [37], [43]. While model-based approaches such as Linear-quadratic Regulator (LQR), model predictive control, and adaptive schemes offer theoretical advantages, the disturbance dynamics considered in this work were sufficiently low-frequency and low-dimensional for a classical PID controller to provide satisfactory stabilisation performance without requiring additional plant-model complexity.

Independent PID controllers were implemented, with the control input $u(t)$ computed according to the standard PID control law:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)\, d\tau + K_d \frac{de(t)}{dt}, \tag{5}$$

where $e(t)$ denotes the attitude error between the desired and measured platform orientation, and K_p, K_i, and K_d are the proportional, integral, and derivative gains, respectively.

7.1.1 Simulation Environment

The simulation environment was implemented in Simscape Multibody due to its direct integration with MATLAB/Simulink and native support for CAD-based multibody dynamics. SolidWorks geometry was imported using the Simscape Multibody Link plugin, preserving the rigid-body mass properties, inertia tensors, and kinematic joint constraints from the mechanical assembly.

The plant model represented the completed CAD assembly, including motor brackets, mounting plates, structural rods, fasteners, and the LiDAR platform. The resulting model comprised 57 distinct revolute joints in total, of which two represent the actively controlled roll and pitch axes. Two additional revolute joints were used to inject recorded chassis disturbances into the simulated plant. The disturbance input dataset was captured during field traversal of the SLaRC UGV. The remaining joints model the structural assembly, primarily representing structural fasteners. A high-level overview of the plant kinematic chain is shown in Figure 6. The complete Simscape Multibody assembly is provided in Appendix D, while a visualisation of the assembled model in Mechanics Explorer is shown in Figure 7.

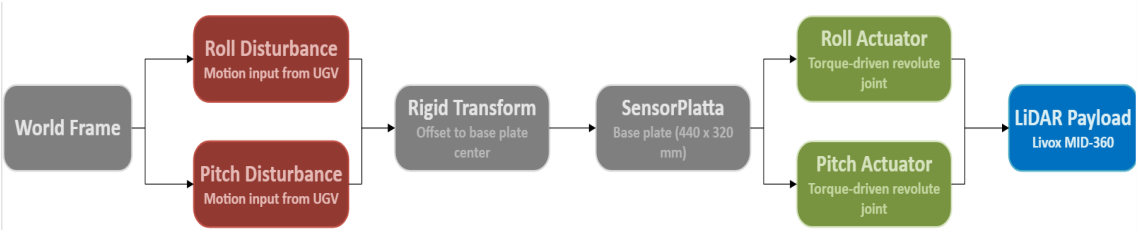


Figure 6: High-level overview of the kinematic chain for the plant. The complete representation is shown in Appendix D.

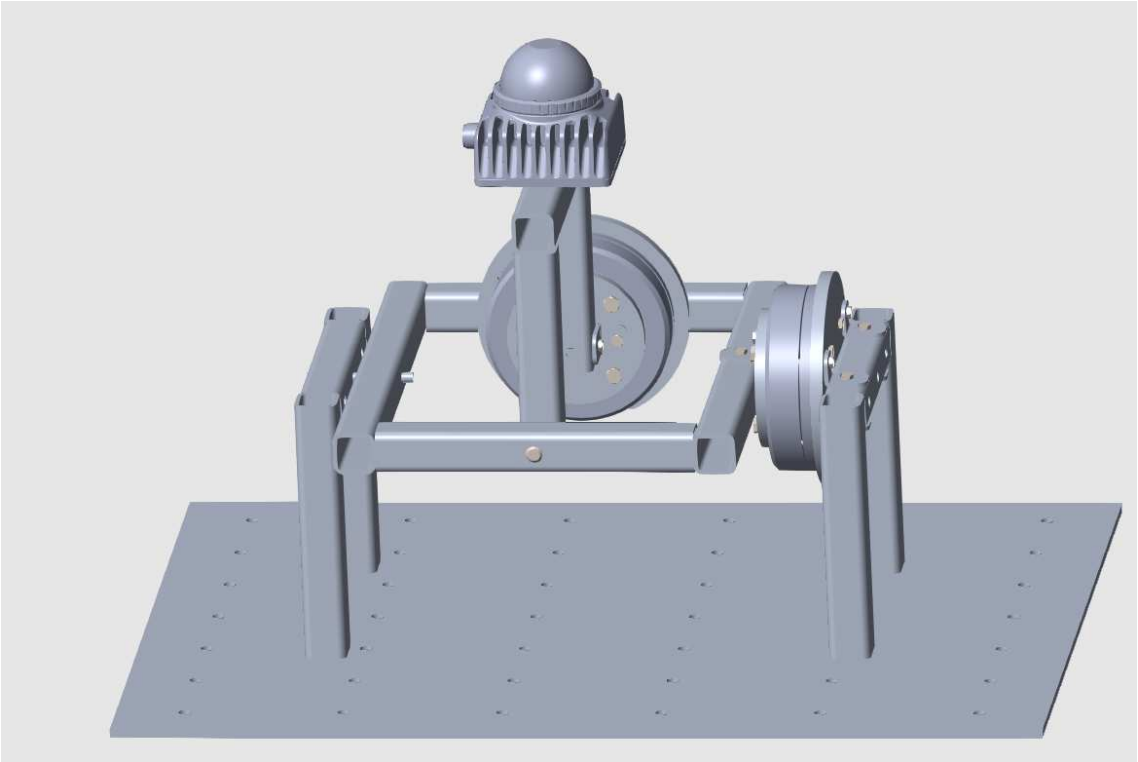


Figure 7: Visualisation of the simulated plant in Simscape Mechanics Explorer at the nominal stabilised configuration.

The world frame defined the global reference coordinate system, with the positive Z-axis oriented vertically upward and gravitational acceleration modelled as defined in Equation 6:

$$\mathbf{g} = \begin{bmatrix} 0 \\ 0 \\ -9.81 \end{bmatrix} \text{ m/s}^2 \tag{6}$$

The imported SolidWorks assembly required coordinate-frame alignment to ensure consistent roll and pitch definitions within the Simscape Multibody environment. A rigid transform was therefore applied between the world and base frame. The corrected configuration yielded zero roll and pitch at the nominal operating point, eliminating the need for controller reference offsets. The gimbal base, represented by the *SensorPlate* rigid body, received disturbance input through two cascaded revolute joints corresponding to roll and pitch chassis motion. The joints operated in motion-input mode, with angular position, velocity, and acceleration supplied directly through Simulink-PS Converter blocks configured for derivative input. Direct derivative input avoided internal numerical differentiation of the disturbance signals and reduced simulation overhead associated with the high-resolution disturbance dataset. The simulation was executed using a variable-step stiff solver

`ode23t`, selected for its suitability for Simscape Multibody systems with constrained multibody dynamics and stiff mechanical interactions. The variable-step formulation additionally reduced computational overhead during low-dynamic intervals while maintaining numerical stability during rapid disturbance transients. The maximum solver step size was constrained to 5×10^{-3} s to maintain consistency with the 200 Hz disturbance dataset. A second rigid transform of $[0.220, 0.160, 0]$ m was inserted between the disturbance input joints and the base body to relocate the disturbance origin to the geometric centre of 440×320 mm base plate.

Above the disturbance input joints, the roll axis was implemented as a revolute joint providing rotational freedom about the roll axis, followed by a second revolute joint governing pitch motion about the orthogonal axis. Both actuated joints operated in torque-input mode, receiving commanded torque from the corresponding PID controller while reporting angular position and velocity feedback. Mechanical end-stops at $\pm 45°$ were modelled using compliant spring-damper elements to prevent unrealistic over-rotation during large disturbance excursions. The LiDAR platform was connected to the terminal link of the kinematic chain.

Representative chassis disturbances were reproduced using a 165 s recording captured during field traversal of the SLaRC UGV. The dataset was recorded as a ROS2 bag using the chassis-mounted 3DM-GV7-AHRS reference IMU at 200 Hz. Quaternion orientation samples were converted to Euler roll and pitch angles, baseline-corrected against the initial interval, and low-pass filtered prior to numerical differentiation, used to obtain the corresponding angular velocity and acceleration channels. The recorded disturbance, shown in Figure 8, exhibited a roll RMS of 6.1° with a peak of 27.9°, and a pitch RMS of 4.0° with a peak of 11.4°, representative of terrain-induced motion during uneven UGV traversal. The resulting disturbance signal, consisting of angular position, velocity, and acceleration channels, was applied directly to the disturbance-input joints as time-varying motion excitation.

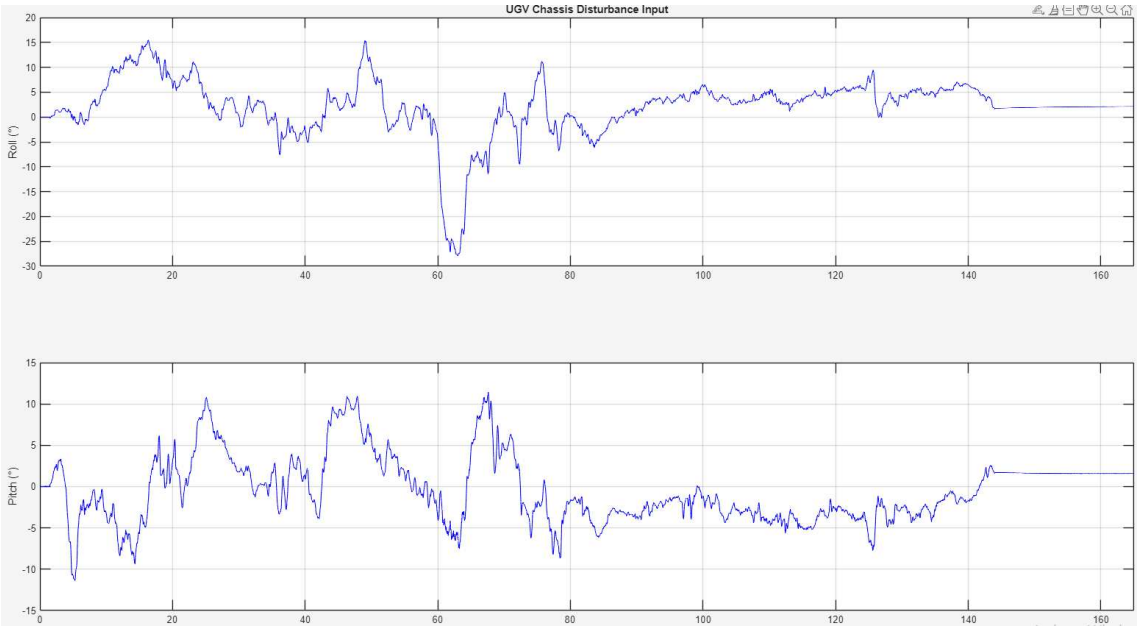


Figure 8: Disturbance input applied to the simulated gimbal base, derived from a 165-second field recording of the SLaRC UGV chassis attitude using the 3DM-GV7-AHRS reference IMU. The dataset was recorded in a Swedish forest, which is the intended operating environment for the SLaRC UGV.

7.1.2 IMU Attitude Extraction

Platform attitude was estimated using a Transform Sensor block in Simscape Multibody, measuring the orientation of the LiDAR payload frame relative to the world frame. The sensor was configured to output the orientation as a sequence of Euler angles using the Z-Y-X intrinsic rotation convention, with the world frame selected as the measurement frame.

The Z-Y-X sequence was selected after empirical evaluation of all six standard Euler-angle decompositions in combination with the corrective Rigid Transform described in Section 7.1.1. With the transform applied, the Z-Y-X decomposition yielded baseline attitude readings within 10^{-3} rad of zero on both controlled axes, whereas alternative sequences placed either the roll or pitch component near $\pm\pi/2$, corresponding to the singular configuration of the respective decomposition, and would have produced discontinuous feedback at smaller disturbances. The selected combination of corrective transform and Z-Y-X decomposition provided a numerically stable feedback signal across the full operating range of $\pm 45°$ required for the stabilisation platform. The Transform Sensor output is a three-element signal corresponding to yaw, pitch, and roll rotations ψ, θ, and ϕ, respectively. A Demux block separated the three channels, where the yaw component was discarded, while the pitch and roll channels were routed to the corresponding control loops.

The first-order low-pass filter form

$$F(s) = \frac{1}{0.01s + 1} \tag{7}$$

was selected as the minimum-complexity approach capable of attenuating high-frequency numerical noise without significantly affecting the low-frequency terrain-induced disturbance content. The filtered roll and pitch signals were then supplied to the corresponding PID controllers as measured process variables. The filter, with a time constant of 10 ms and cut-off frequency of approximately 16 Hz, suppressed solver-induced high-frequency numerical noise and prevented amplification of transient artifacts within the PID derivative action, while preserving the disturbance bandwidth associated with uneven terrain UGV traversal.

7.1.3 PID Control Architecture

The controller was designed to address four practical requirements specific to the SLaRC UGV application. First, the system must attenuate low-frequency disturbances spanning approximately 0.1 to 5 Hz, representing the dominant disturbance band during uneven-terrain UGV traversal as identified in the spectral analysis presented in Section 8. Second, the controller must compensate for the persistent gravitational moment arising from the asymmetric mass distribution about the roll axis, which produces a non-zero steady-state attitude offset under proportional control alone. Third, the controller must remain practically tunable despite modelling uncertainties arising from actuator behaviour and mechanical dynamics. Lastly, the feedback signal must reference the world frame to maintain a gravity-aligned LiDAR orientation during terrain traversal.

Ahmad et al. [37] report effective disturbance rejection for comparable 2-DOF stabilisation platforms under varying base angular motion, while Ghaeminezhad et al. [43] establish PID control as a sufficient baseline for evaluating more advanced control extensions. PID control additionally provides straightforward integration with anti-windup mechanisms and actuator saturation limits, which were required to constrain the commanded torque within the rated capacity of the CubeMars GL80 actuator.

The control input $u(t)$ for each axis was computed according to the standard PID control law, represented in Equation 5. The attitude error $e(t)$ was defined as the difference between the desired platform orientation, set to zero radians for both axes, corresponding to the nominal upright platform orientation, and the measured orientation derived from the Transform Sensor described in Section 7.1.2. The control architecture for a single axis is illustrated in Figure 9. The error signal $e(t)$ was fed simultaneously to three parallel branches: a proportional gain Kp, an integrator with gain Ki, and a filtered derivative term with gain Kd. The derivative action was implemented as a filtered differentiator of the form:

$$D(s) = K_d \cdot \frac{Ns}{s + N} \tag{8}$$

where N denotes the derivative filter coefficient limiting high-frequency amplification of measurement noise inherent to numerical differentiation. A feedforward branch was included in the controller structure but disabled during simulations presented in this work by setting its gain to zero. Closed-loop feedback provided sufficient disturbance rejection within the evaluated operating regime, while the dominant pitch-axis limitation originated primarily from mechanical resonance

rather than insufficient controller responsiveness. Consequently, disturbance rejection was performed exclusively through closed-loop PID feedback. The PID controllers operated in continuous time within the Simulink environment, inheriting the variable solver step size described in Section 7.1.1. The combined control signal was constrained to the actuator torque limits ±2.9 Nm, corresponding to the peak rated torque of the CubeMars GL80 actuator described in Section 7.2.1.

The integrator was independently bounded to ±1.0 N m to prevent wind-up during periods of motor saturation. The remaining ±1.9 N m of commanded torque was reserved for the proportional and derivative branches. This anti-wind-up configuration prevents the integral term from accumulating beyond what the actuator can physically deliver, ensuring rapid recovery following actuator desaturation. Furthermore, a unity-gain inverting block with negative gain equal to -1, was inserted between the saturation output and the actuator input on both axes. This sign correction was determined by verifying that a positive attitude error produced a restoring torque acting in the correct direction. The inversion compensated for the sign-convention mismatch between the controller output and the Simscape revolute joint torque direction defined by the imported CAD model.

The PID gains used in the simulations were tuned empirically through iterative closed-loop response evaluation across the complete disturbance dataset. The selected values are summarised in Table 1

Axis	K_p	K_i	K_d	Saturation (N·m)
Roll	25	0.1	3	±2.9
Pitch	50	1.0	10	±2.9

Table 1: Selected PID gains and actuator saturation limits used in the simulation model.

The controllers operated independently in the feedback architecture, with each axis receiving only its corresponding Transform Sensor-derived attitude signal. The Transform Sensor reported each angle in the world reference frame rather than relative to the immediate parent body. Combined with the small residual angles maintained by the closed-loop stabilisation, typically below 1° during steady-state operation, this allowed the roll and pitch axes to be treated as independent single-input single-output control loops within the intended operating range. Consequently, pitch disturbances induced by roll motion were reflected directly in the pitch feedback signal without requiring explicit cross-axis compensation. However, mechanical cross-axis coupling between the roll and pitch axes persisted due to the serial gimbal geometry, where rotational acceleration about one axis transmits forces through the supporting structure to the other.

The motor command output from each controller was passed to the Simscape revolute joint via a conversion gain that maps the normalised control signal to a torque value in N m.

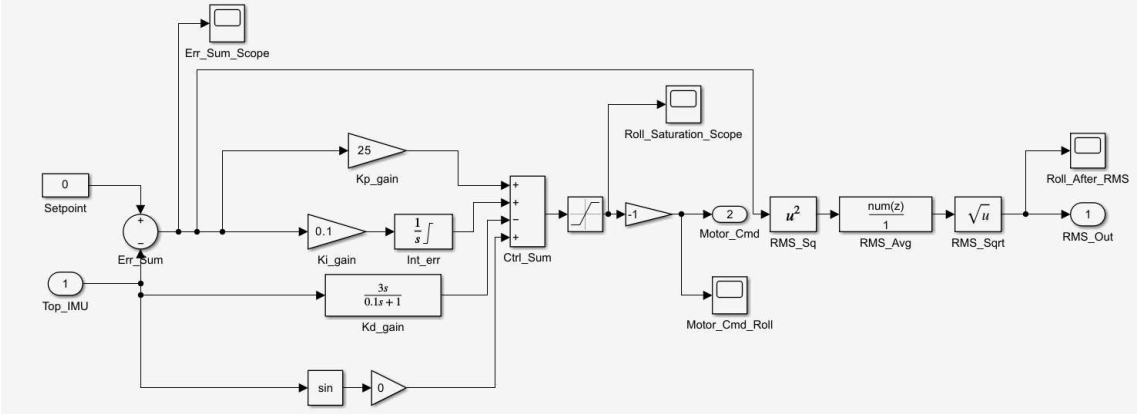


Figure 9: Block diagram of the PID control architecture for a single stabilisation axis, in this case roll. The structure is identical for both roll and pitch controllers except the parameter values.

7.1.4 Stability Detection

Stability detection was introduced to quantify the duration for which the stabilised platform remained within the attitude-error bounds required for perception-relevant LiDAR operation. A stability detection mechanism was implemented for the roll and pitch axes using Transform Sensor-derived attitude signals to identify intervals suitable for reliable LiDAR acquisition. As the detection pipeline is structurally identical for both axes, only a single-axis implementation is described herein, with axis-specific threshold values presented in Table 2.

The attitude signal for each axis was processed using a moving RMS operation evaluated over a sliding 50 ms window corresponding to 50 samples at the 1 kHz filter rate. The RMS operation was implemented as a uniform-weighted Finite Impulse Response (FIR) filter applied to the squared attitude signal, followed by a square-root operation, quantifying the sustained magnitude of platform motion independently of rotation direction. The resulting RMS signal was supplied to an Set-Reset (SR) latch governing the binary stability state of each axis. The SR latch was set to a stable state when the RMS value fell below a lower threshold ϵ_{set} defined in Table 2, indicating that platform motion had attenuated below the required stability limit. Stability was invalidated when the RMS value exceeded the upper threshold ϵ_{reset}, providing hysteretic separation between stability assertion and reset conditions. This hysteretic behaviour prevented spurious toggling of the stability signal due to marginal or transient motion near a single threshold boundary.

Axis	$\boldsymbol{\epsilon}_{\text{set}}$ **(rad)**	$\boldsymbol{\epsilon}_{\text{reset}}$ **(rad)**
Roll	0.001	0.01
Pitch	0.001	0.01

Table 2: Stability detection thresholds for the roll and pitch axes.

The thresholds $\boldsymbol{\epsilon}_{\text{set}}$ and $\boldsymbol{\epsilon}_{\text{reset}}$ were selected empirically based on the noise floor observed in the IMU reference measurement discussed later in Section 7.4.1, with the upper threshold providing a $5\times$ margin over the lower to ensure robust hysteretic behaviour. The individual stability outputs of the roll and pitch SR latches, denoted as S_{roll} and S_{pitch}, were combined using a logical AND operation, as illustrated in Figure 10. The resulting signal, S_{stable}, represented the global stability condition of the platform and was asserted only when both axes simultaneously satisfy their respective stability criteria. LiDAR measurements acquired while S_{stable} was asserted were classified as valid for subsequent geometric evaluation.

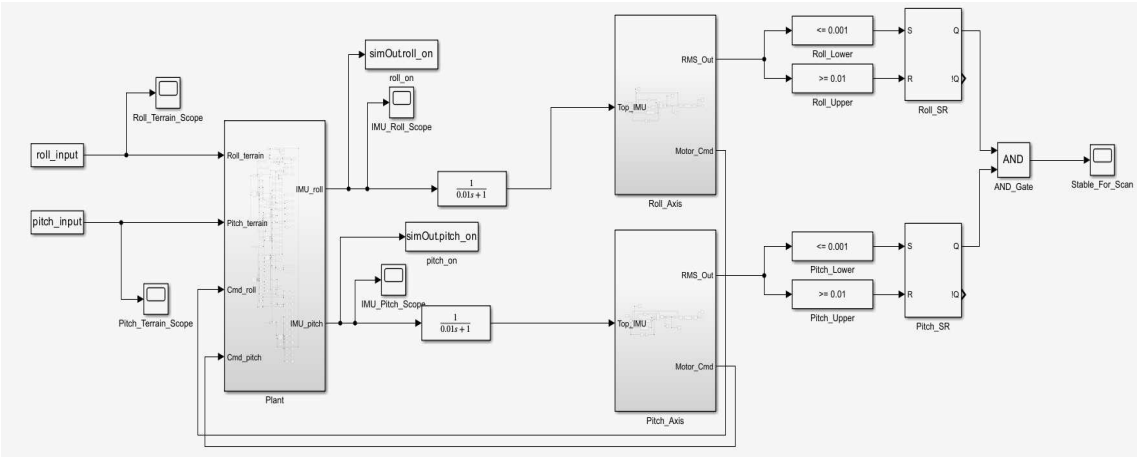


Figure 10: Block diagram of the main controller including the roll-axis, pitch-axis and plant, along with the SR latch.

7.2 Design

This subsection describes the hardware components selected for the physical realisation of the actively stabilised sensor platform. The design decisions were based on commercially available components and real-world dimensions to maintain physically representative simulation results. Furthermore, the use of existing components simplifies the future transition from simulation to physical implementation. Component selection was guided by the torque, bandwidth, and sensing accuracy requirements derived in Section 5.4, together with practical constraints imposed by the SLaRC UGV platform, including available supply voltage and mechanical integration constraints. The selected hardware components for actuation, motor control, and attitude sensing are described in the following subsections.

7.2.1 Actuation System

The actuation system was designed to provide sufficient torque and dynamic response to stabilise the LiDAR platform under terrain-induced roll and pitch disturbances. The following subsection describes the actuator requirements, motor selection, and controller integration used in the proposed stabilisation platform.

Actuation Requirements: The stabilisation platform requires active actuation along both the roll and pitch rotational axes to counteract terrain-induced disturbances and gravitational loading while maintaining sensor attitude stability. Each axis demands an actuator delivering sufficient torque to maintain the top platform within the required stability threshold of ± 0.01 rad ($\pm 0.57°$), even under dynamic loading conditions.

The passive stabilisation mechanism present on the SLaRC UGV was designed to absorb shocks and sudden movements, but does not provide attitude stabilisation for the sensor payload. Active motorised actuation is therefore required to enable closed-loop feedback control, where motor torque is commanded in real-time based on IMU feedback. The actuator must respond with sufficient bandwidth and available torque to reject disturbances before they propagate to the payload.

Actuator Selection: The required actuator torque was derived from the static and dynamic loading conditions of the platform. The top platform, including the LiDAR payload, has an estimated mass of $m = 1.5$ kg, with the centre of mass located at a distance of $r = 0.09$ m from the gimbal rotation axis.

The gravitational torque acting on the platform at maximum operating tilt angle $\theta_{\max} = 30$ is:

$$\tau_{\text{grav}} = m \cdot g \cdot r \cdot \sin(\theta_{\max}) = 2.125 \cdot 9.81 \cdot 0.09 \cdot \sin(30) \approx 0.938 \text{ Nm} \tag{9}$$

To account for dynamic effects during terrain disturbances, an inertial torque component is added. The moment of inertia at the top platform about the gimbal axis is estimated as:

$$J = \frac{1}{12} \cdot m \cdot (a^2 + b^2) = \frac{1}{12} \cdot 2.125 \cdot (0.18^2 + 0.18^2) \approx 0.0115 \text{ kg·m}^2 \tag{10}$$

Assuming a maximum angular acceleration of $\alpha = 5$ rad/s^2 during active stabilisation:

$$\tau_{\text{inertia}} = J \cdot \alpha = 0.0115 \cdot 5 \approx 0.057 \text{ Nm} \tag{11}$$

Including a safety factor of $k = 2$ to account for model uncertainties, friction, and peak disturbance loads, the required peak actuator torque is:

$$\tau_{\text{req}} = k \cdot (\tau_{\text{grav}} + \tau_{\text{inertia}}) = 2 \cdot (0.938 + 0.057) \approx 1.99 \,\text{Nm} \tag{12}$$

Continuous operation over extended terrain traversal additionally requires a duty cycle of 100 %. The required torque calculated in Equation 12 and the continuous uptime make the majority of linear actuators insufficient for this application. Stepper motors were considered as an alternative due to their inherent position control capability and low cost. However, they present several fundamental limitations for this thesis. Stepper motors were considered as an alternative due to

their inherent position control capability and low cost. However, their suitability and commercial availability for continuous high-dynamic stabilisation tasks under varying terrain-induced loading conditions were considered limited for the intended application. Furthermore, they are not designed for continuous duty cycle operation at high torque heat build-up, necessitating torque derating to prevent thermal damage. Finally, the inherent cogging torque of stepper motors introduces torque ripple that would degrade stabilisation performance at low angular velocities [65].

The CubeMars GL80 frameless direct-drive actuator was selected based on the torque requirements derived in Equations 9–12 and its compatibility with the SLaRC UGV 24 V power system. The GL80 provides a peak torque of 2.9 Nm and a nominal torque of 1.405 Nm, while also having a no-load speed of 650 RPM and a rated speed of 450 RPM, yielding sufficient torque over the calculated requirement, as well as sufficiently high rotational speed to accommodate for quick fluctuations caused by the terrain disturbances. The GL80 operates as a direct-drive actuator, reducing gearbox-induced backlash and non-linearities that would otherwise degrade closed-loop control performance. Two actuator units were selected for the proposed platform architecture, one per rotational axis, mounted coaxially with the roll and pitch gimbal axes, respectively.

Motor Controller Integration: Each GL80 actuator is intended to be controlled using a dedicated Moteus R4.11 brushless motor controller. The Moteus R4.11 was selected for its native support of CAN-FD communication, enabling high-bandwidth and low-latency torque command transmission between the onboard ASUS NUC 13 and the motor controllers. The controller internally implements field-oriented control, allowing the stabilisation controller to operate at the torque-command level without requiring external current-loop implementation. The proposed CAN-FD topology connects the onboard computer to both motor controllers through a shared communication bus, as illustrated in Figure 11.

Fault Handling and Protection: The open-source firmware of the Moteus controller allows configurable current limits, velocity limits, and fault-handling behaviour, supporting integration within the proposed stabilisation framework.

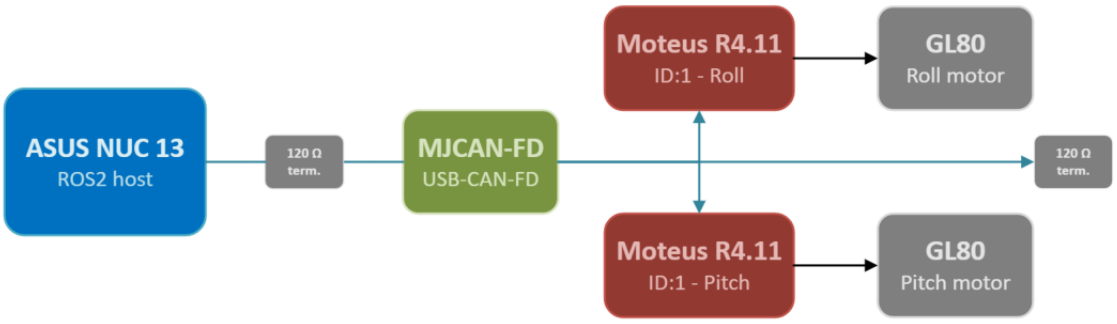


Figure 11: CAN-FD bus topology connecting the ASUS NUC 13 to both Moteus R4.11 motor controllers via the MJCAN FD USB-CAN-FD adapter. The MJCAN FD provides integrated bus termination at the host end, with a single external 120 Ω resistor placed at the distal end of the bus after the final Moteus controller.

7.2.2 Inertial Measurement Unit

The IMU selected for the proposed stabilisation platform is a tactical-grade Attitude and Heading Reference System (AHRS), incorporating a triaxial accelerometer, gyroscope, and magnetometer. The 3DM-GV7-AHRS was selected due to its integrated real-time attitude estimation, tactical-grade orientation stability, and robustness. All of which are crucial to maintain high-accuracy long-range LiDAR geometric consistency during uneven terrain-traversal, where the acceptable deviation threshold is ±0.01 radians (0.57°).

The gyroscope bias instability of 1.5 °/hr and angular random walk of 0.3 °/$\sqrt{\text{hr}}$ were considered sufficient to prevent accumulated attitude drift from exceeding the required stability threshold

during continuous operation [66]. Similarly, the accelerometer bias instability of 20 μg and random walk of 20 $\mu g/\sqrt{\text{Hz}}$ were considered sufficient for reliable tilt estimation.

The sensor outputs fused attitude estimates via an onboard Extended Kalman Filter (EKF) at an update rate of 1000 Hz, which is well above the bandwidth required for the stabilisation controller. The reported roll and pitch accuracy of 0.25° static and 0.5° dynamic is within the required stability threshold, supporting its suitability for closed-loop stabilisation.

The 3DM-GV7-AHRS was primarily used for chassis attitude acquisition during the recorded UGV traversal dataset later discussed in Section 7.4.1. The IMU was mounted directly within the UGV chassis to capture terrain-induced disturbances prior to stabilisation, enabling representative disturbance input separate from the internal LiDAR IMU measurements used within the stabilised sensor platform. The sensor supports ROS-compatible communication interfaces, facilitating integration with the SLaRC UGV software architecture. Additionally, the IP68 ingress protection rating and operating temperature range of -40°C to 85°C support reliable operation in outdoor terrain environments.

7.3 Mechanical Construction

The gimbal structure was designed in SolidWorks and underwent multiple design iterations prior to the final configuration shown in Figure 12. The design was developed to remain mechanically realisable using commercially available components and standard aluminium profiles for structural construction.

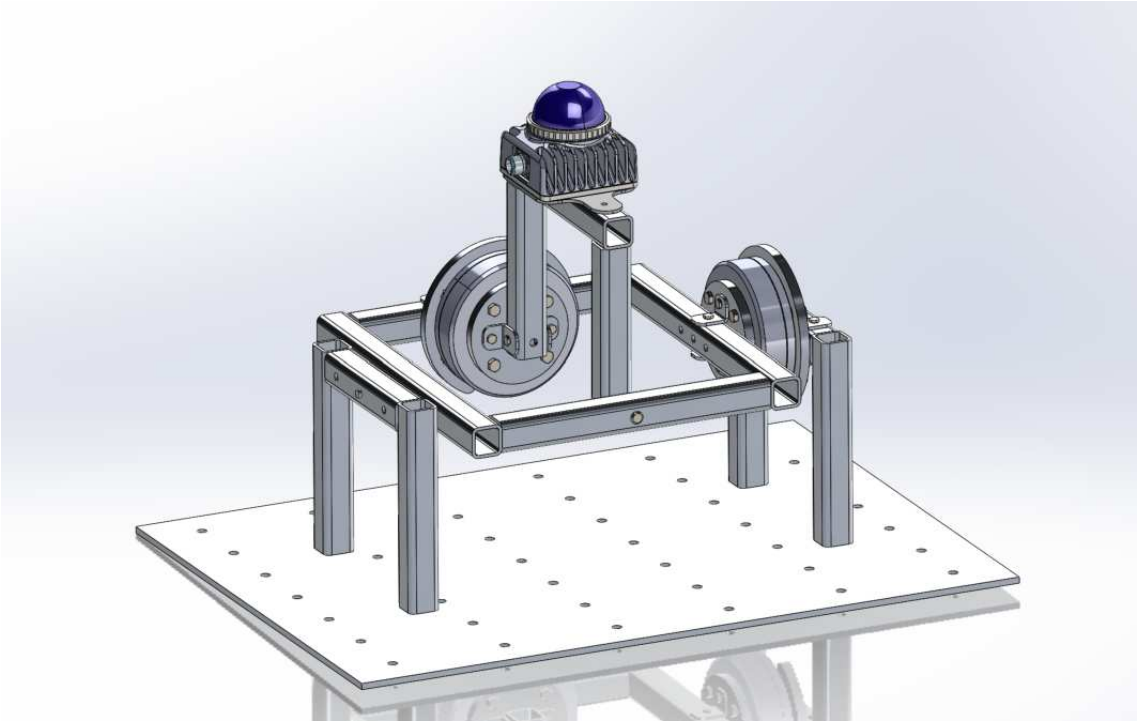


Figure 12: Finalised CAD assembly of the gimbal stabilisation platform.

An initial concept under consideration was a Stewart platform, a parallel manipulator capable of full six-degree-of-freedom motion control [67]. While theoretically offering superior disturbance rejection, this approach was discarded due to its substantially higher control complexity, the additional power infrastructure of the SLaRC UGV required for six actuators, and the limited benefit of full six-degree-of-freedom stabilisation for the roll- and pitch-dominated disturbances considered in this thesis.

The primary structural material was a 20×20 mm aluminium profile, selected for its favourable stiffness-to-weight ratio and modular assembly characteristics, allowing iterative redesign without

requiring complete structural reconstruction. The structural platform has a mass of 1230 g excluding motors and sensors. Each actuator contributes an additional 315 g, while the Livox MID-360 contributes 265 g, resulting in a total platform mass of approximately 2125 g excluding fasteners.

The final design implements a serial two-axis kinematic chain consistent with the Simscape model described in Section 7.1.1. The outer stage consists of four vertical support legs, two on each side, connected by a horizontal crossbar, forming the roll frame. The CubeMars GL80 roll actuator is mounted coaxially with this crossbar, driving rotation about the roll axis. Mounted centrally on the crossbar is a vertical bracket forming the inner pitch stage, where the second GL80 actuator drives rotation about the pitch axis. The actuators were mounted directly about their respective rotational axes to simplify the mechanical transmission path and reduce additional backlash or structural compliance within the stabilisation mechanism. The Livox MID-360 LiDAR is mounted on top of the platform at the terminal link of the kinematic chain. The Livox unit contains an internal IMU that directly measures the orientation of the sensor payload, intended to provide the feedback signal for the proposed closed-loop stabilisation framework as described in Section 7.2.2. A second IMU, the 3DM-GV7-AHRS, is mounted inside the SLaRC UGV chassis to measure the disturbance input to the stabilisation system. This dual-IMU configuration enabled direct comparison between chassis motion measured by the 3DM-GV7-AHRS and corresponding stabilised platform motion measured by the Livox internal IMU, allowing quantitative evaluation of closed-loop disturbance rejection performance. Motor-to-profile interfaces were realised using 6 mm aluminium mounting plates secured using bent 2 mm aluminium brackets and mechanical fasteners.

7.4 Experiments

To evaluate the performance of the simulated active stabilisation system under realistic operating conditions, a structured experimental test was conducted on the SLaRC UGV. The evaluation assessed the ability of the 2-DOF gimbal to maintain a near-horizontal sensor orientation across a range of terrain and motion profiles. Since the IMU-derived attitude angles ϕ and θ, representing roll and pitch deviation relative to the world frame, directly correspond to the LiDAR sensor pointing error, IMU-based metrics constituted the primary quantitative measure of stabilisation performance. Point cloud analysis was additionally performed to provide empirical evidence of the relationship between attitude error and LiDAR geometric consistency, directly addressing RQ3.

7.4.1 Test Environment and Data Acquisition

To address the research questions presented in Section 2., two complementary datasets were required. The first captured terrain-induced UGV attitude disturbances for stabilisation evaluation under realistic conditions, while the second characterised LiDAR geometric behaviour under controller rotational motion for point cloud distortion analysis.

Two independent datasets were acquired for the stabilisation and point-cloud evaluation pipeline. The first dataset consisted of a 165 s terrain traversal recording captured using the SLaRC UGV operating on uneven outdoor terrain, more specifically, in a Swedish forest. The chassis attitude was measured using the 3DM-GV7-AHRS IMU mounted inside the UGV chassis and sampled at $200, \mathrm{Hz}$. Simultaneously, the Livox MID-360 mounted on the sensor platform recorded LiDAR data together with its internal IMU. All sensor streams were captured within a shared ROS2 bag file, providing temporal alignment. The relevant ROS2 topics were extracted using FoxGlove Studios and exported to Comma-Separated Values (CSV) format for subsequent processing. Only the chassis-mounted IMU attitude signals from this recording were used in the present analysis, where the recorded roll and pitch time series served as disturbance input to the simulation described in Section 7.1. The signal-processing pipeline used for the extraction and post-processing of the recorded IMU data is illustrated in Figure 13.



Figure 13: Signal processing pipeline for IMU attitude data, from raw ROS2 bag recording to metric computation in MATLAB.

The recorded angular velocity and orientation signals were processed to establish the expected disturbance amplitudes and dynamic behaviour relevant to the stabilisation controller. The resulting measurements are illustrated in Figure 14.

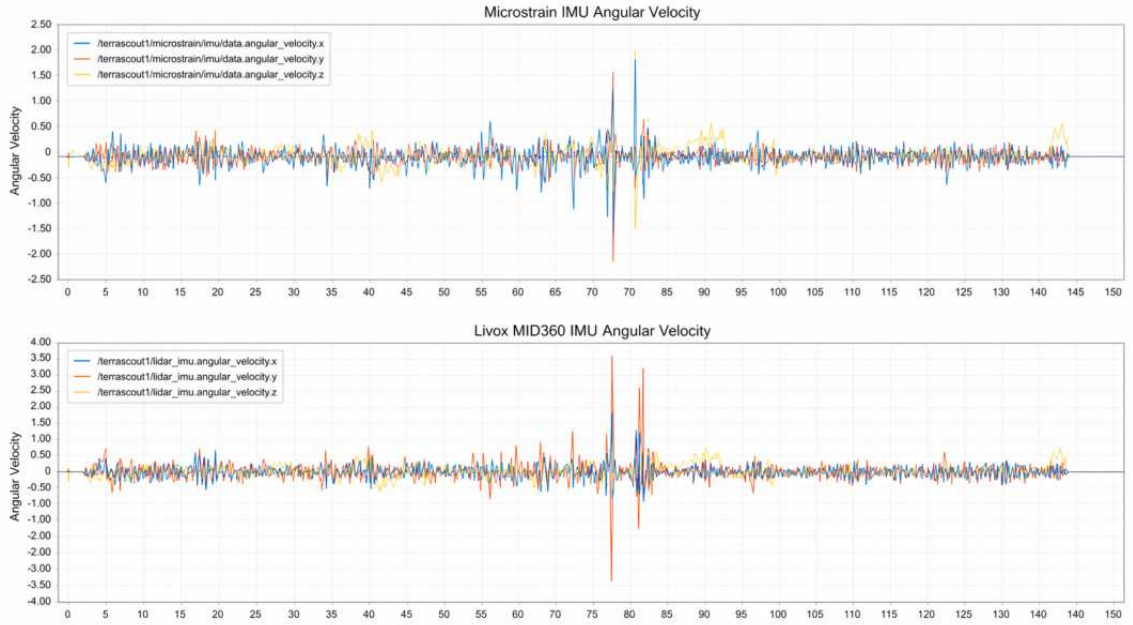
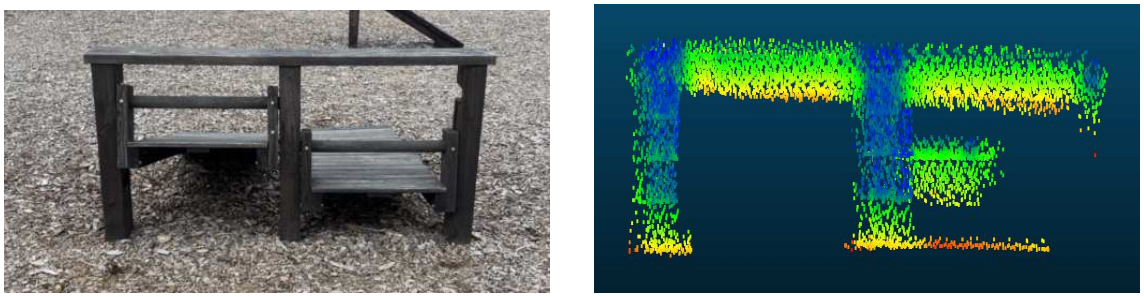


Figure 14: Angular velocity comparison between chassis-mounted 3DM-GV7-AHRS (above) and platform-mounted Livox MID-360 internal IMU (below) during baseline UGV traversal. The platform-mounted sensor experiences peak angular velocities of ±3.5 rad/s compared to ±2.0 rad/s on the chassis, indicating mechanical amplification that motivates active stabilisation.

The second recording consisted of a dynamic reference experiment acquired using the Livox MID-360 in an outdoor gym environment to characterise the empirical LiDAR measurement properties of a planar reference scene. The sensor was mounted to an aluminium profile through a custom 3D-printed mount permitting controlled manual rotation about the roll and pitch axes. The acquisition consisted of dynamic manual tilt motions spanning approximately ±30° were made, controlled by a gyroscope on a mobile device parallel with the aluminium profile. A planar wooden bench with the width 1.73 m and thickness 0.042 m was positioned at an approximate distance of 2 m from the LiDAR and served as the stationary reference object throughout the recording and the corresponding point cloud, which is shown in Figure 15a and Figure 15b respectively.



(a) Physical Reference (b) Corresponding Point Cloud

Figure 15: LiDAR reference setup for controlled roll and pitch excitation during point cloud acquisition and the corresponding point cloud after the Random Sample Consensus (RANSAC) implementation.

Data acquisition was performed using the LivoxViewer software and exported directly as CSV point cloud data containing (X, Y, Z, t) tuples.

The two recordings served complementary purposes within the evaluation framework. The terrain traversal recording provided the disturbance input used throughout the stabilisation simulation pipeline, while the reference recording provided both the empirical σ_{LiDAR} characterisation through RANSAC-based plane fitting and the baseline reference cloud used for the synthetic point cloud comparison described in Section 7.4.3.

7.4.2 Point Cloud Analysis

Point cloud analysis was performed to establish a quantitative relationship between residual stabilisation error and perception-relevant geometric distortion in the reconstructed LiDAR measurements. Since the research questions address both angular disturbance rejection and geometric point cloud consistency, the analysis pipeline was designed to estimate the intrinsic LiDAR measurement noise together with the additional range-dependent error introduced by residual platform motion. Plane-based analysis using RANSAC segmentation was selected due to its robustness against outliers and its suitability for extracting repeatable geometric reference features from noisy point cloud measurements.

A custom Python pipeline was implemented to characterise the LiDAR measurement properties from the controlled reference recording described in Section 7.4.1 and to translate residual platform attitude error into a range-dependent measurement uncertainty. The pipeline extracted CSV exports from the LivoxViewer software and was structured as four sequential stages: data loading and filtering, plane segmentation, temporal angular analysis, and error budget computation. The overall architecture is shown in Figure 16.

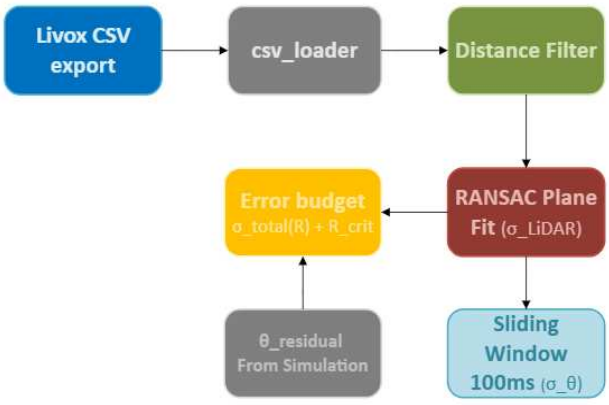


Figure 16: Architecture of the LiDAR analysis pipeline.

The simulated residual attitude signals produced by the stabilisation model were combined with the measured LiDAR noise floor through this pipeline in order to produce the range-dependent uncertainty model reported in Section 8.

The data loading stage parsed the CSV export, removed invalid returns, computed spherical distances for each point, and applied a distance-window filter to retain candidate points near the reference object described in Section 7.4.1 as a wooden bench. Plane segmentation was performed using RANSAC [68] on the filtered candidate set, using 200 iterations, an inlier threshold of 3 cm, followed by least-squares refinement of the recovered plane parameters using the identified inlier set. The output consisted of a unit normal, the plane centroid, the inlier mask, and the plane-fitting residual RMS. The resulting plane-fitting residual RMS, evaluated during the reference interval, was used as the empirical estimate of the LiDAR measurement noise σ_{LiDAR}. An illustration of the segmentation result on the reference recording is shown in Figure 17.

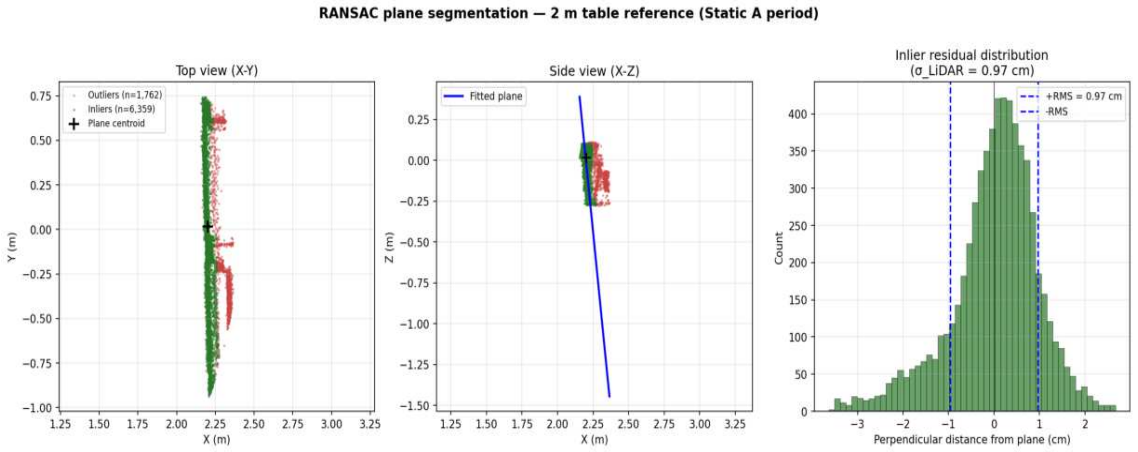


Figure 17: The top view (left) shows inliers (green) correctly identified as the bench surface and outliers (red) including supporting structure. The side view (centre) shows the fitted plane intersecting the inlier cluster. The right panel shows the perpendicular distance distribution of inliers, from which σ_{LiDAR} is derived.

Temporal analysis was performed by applying the same RANSAC segmentation procedure within a sliding window of 100 ms duration. This produced a time series of plane normals from which the angular deviation σ_θ was computed as the standard deviation of the angle between successive plane normals and a reference plane normal taken from the first valid window. Sign ambiguity in the recovered normal direction was resolved by aligning each window's normal with the reference, characterising the angular fluctuation of the sensor pose during the controlled data acquisition. The sliding-window concept and representative output for the reference period are illustrated in Figure 18. The σ_{LiDAR} from the segmentation stage and the simulated residual angular error $\theta_{\mathrm{residual}}$ produced by the stabilisation model were combined into a range-dependent error budget. The total vertical measurement uncertainty at a range R was modelled as the quadrature sum of the LiDAR noise and the angular projection of platform attitude error onto the measurement direction:

$$\sigma_{\mathrm{total}}(R) = \sqrt{\sigma_{\mathrm{LiDAR}}^2 + \left(R \cdot \sin\theta_{\mathrm{residual}}\right)^2} \tag{13}$$

The critical range R_{crit}, defined as the range at which the angular contribution equals the LiDAR noise, was obtained as

$$R_{\mathrm{crit}} = \frac{\sigma_{\mathrm{LiDAR}}}{\sin\theta_{\mathrm{residual}}} \tag{14}$$

To characterise the closed-loop stabilisation behaviour as a function of frequency, the Power Spectral Density (PSD) of the chassis disturbance input and the residual stabilised output was estimated using Welch's method [69]. The chassis attitude derived from the terrain traversal recording described in Section 7.4.1 together with the simulated platform attitude was first resampled onto a uniform 200 Hz time base via interpolation, after which each signal was centred about its mean.

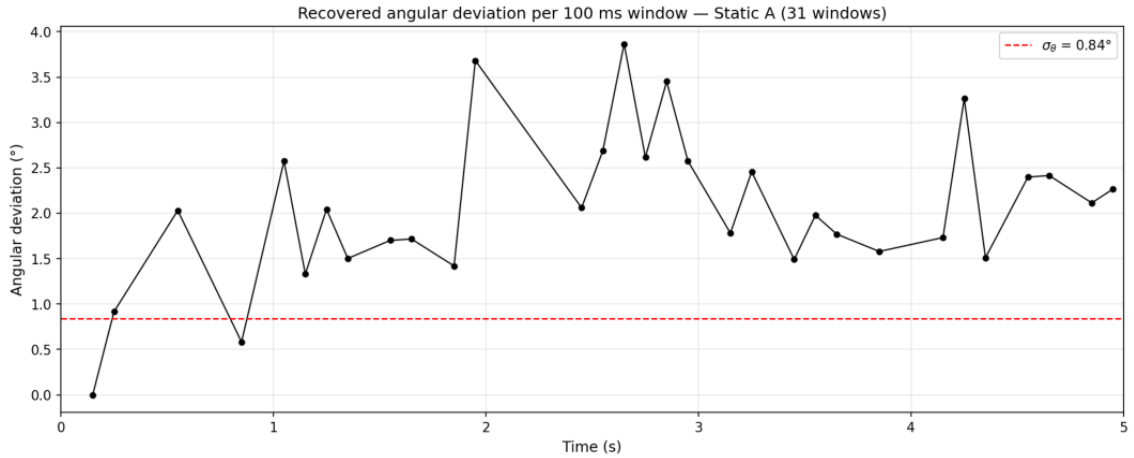


Figure 18: A 100 ms window slides over the recording (top), and each window position produces a plane normal$\mathbf{n}_i$ via independent RANSAC segmentation. The angular deviation between successive normals and a reference normal is recovered as a time series (bottom), with σ_θ defined as its standard deviation across the static reference period.

Each PSD was then computed using a Hann window of 2048 samples, 50% overlap, and 4096 Fast-Fourier Transform (FFT) points, yielding a frequency resolution of approximately 0.05 Hz over the analysis band.

Per-frequency disturbance attenuation was computed as

$$A(f) = 10 \log_{10}\left(\frac{P_{\text{off}}(f)}{P_{\text{on}}(f)}\right) \quad [\text{dB}], \tag{15}$$

where P_{off} and P_{on} denote the PSDs of the unstabilised chassis disturbance and the stabilised platform output, respectively. Positive values of $A(f)$ indicate disturbance rejection by the closed loop, whereas negative values indicate amplification of disturbance energy by the closed loop.

7.4.3 Point Cloud Comparison

Point cloud comparison was performed to quantify the geometric distortion introduced by residual platform motion in reconstructed LiDAR measurements. Synthetic point clouds generated from the recorded chassis attitude and simulated stabilised residuals were compared against a reference scene to evaluate whether active stabilisation improves geometric consistency relative to rigid mounting. The selected metrics provide complementary measures of distortion, where ICP evaluates global registration consistency, C2C quantifies local geometric displacement, and M3C2 estimates surface-normal-aware geometric deviation. The selected comparison algorithms represent established methods for geometric point cloud analysis and are widely used in software frameworks such as `CloudCompare` [70].

The LiDAR reference dataset described in Section 7.4.1 was rotated point-by-point according to two attitude profiles: the recorded UGV chassis motion and the simulated platform residual platform motion produced by the closed-loop Simscape model. Both synthetic clouds represented the same reference scene as observed by an LiDAR subjected to the respective attitude profiles. They were subsequently compared against the baseline reference cloud to quantify the geometric deviation introduced by each attitude profile.

For each point in the baseline reference cloud, an attitude rotation was applied about the sensor origin using the roll and pitch values interpolated to that point's individual timestamp. The Z-Y-X convention is consistent with the simulation plant used:

$$\mathbf{R}(\phi, \theta) = \mathbf{R}_y(\theta)\, \mathbf{R}_x(\phi), \tag{16}$$

where ϕ and θ represent roll and pitch, respectively. A 4-second window of the full SLaRC UGV recording spanning $t = 50\,\text{s}$ to $54\,\text{s}$ was selected as the disturbance input with roll and pitch

RMS values matching the aggregate full-recording statistics within 15 %. The baseline cloud was generated by applying the unfiltered UGV recording attitude to each reference point. The stabilised cloud was generated by applying the corresponding platform residual to the same set of reference points using identical timestamp interpolation for both attitude profiles.

A custom Python implementation of the cloud comparison methodology was then developed, drawing on algorithms established in the `CloudCompare` software [70]. The implementation comprised three components: ICP registration, C2C nearest-neighbour distance, and M3C2 [24]. Surface normals required for both ICP and M3C2 were estimated via principal component analysis on the $k = 10$ nearest neighbours of each point, with consistent orientation enforced relative to the sensor origin.

ICP registration was implemented as a point-to-plane variant using small-angle linearisation of the rotation matrix, where the linearised least-squares problem was solved iteratively with the incremental transformation accumulated after each iteration. The convergence tolerance was set to 10^{-6} on RMSE change between successive iterations and the maximum iteration count to 20, matching standard `CloudCompare` defaults.

For *C2C distance*, the Euclidean distance between each point in the test cloud and its nearest neighbour in the reference cloud was computed using a k-d tree spatial index. Summary statistics, including mean distance, median distance, RMS distance, and the 95th and 99th percentiles of the resulting distance distribution, were computed to characterise both typical and worst-case geometric deviation.

For each test point in *M3C2 distance*, a cylinder of diameter $D = 0.10\,\text{m}$ and projection depth 1.00 m was constructed along the locally estimated normal direction. The signed distance was computed as the mean projection of reference points falling within the cylinder onto the test-point normal. The cylinder diameter was selected to match the centimetre-scale geometry of the planar reference scene. The 2.00 m default in `CloudCompare` is usually intended for terrain-scale analyses and was found unsuitable for this particular scene. The level of detection at 95 % confidence was computed as

$$\text{LOD}_{95} = 1.96\sqrt{\sigma_{\text{test}}^2 + \sigma_{\text{ref}}^2} + \text{RMSE}_{\text{ICP}}, \tag{17}$$

propagating the registration uncertainty into the change-detection threshold. Points for which the absolute M3C2 distance exceeds LOD_{95} are classified as having undergone significant geometric change. Points without sufficient reference neighbours within the cylinder are excluded from the analysis and classified as outliers. The proportion of valid points additionally served as a measure of cloud overlap, since severely displaced clouds fail to sufficiently produce local correspondences.

8. Results

This section presents the results obtained from the stabilisation simulation using the IMU attitude data recorded during real traversal of the SLaRC UGV as terrain disturbance input. The simulation evaluates the closed-loop stabilisation performance in both roll and pitch. The validation is structured around four connected analyses. First, the simulation was evaluated in both the time and frequency domains to quantify disturbance rejection. Second, the stability detection mechanism was evaluated across the full 165 s simulation interval using the stability thresholds defined in Section 7.1.4. Third, the residual angular errors were evaluated together with the range-dependent error budget defined in Equation (13). Lastly, the simulated residuals were applied to a synthetic point cloud experiment for comparison against the rigidly mounted baseline.

The recorded UGV disturbance described in Section 7.1.1 was applied to the closed-loop plant of Figure 9 over the full 165-second simulation interval. The resulting residual attitude values are summarised in Table 3. The measured roll DRR was 59.9×, while the measured pitch DRR was 7.8×, where DRR is defined as the ratio of input RMS to output RMS. Furthermore, the maximum measured pitch residual was 6.347°. A slow transient response is observed in the roll during the first approximately 60 s of the simulation, shown in Figure 19. The bias is reduced from approximately 0.40° at $t = 0$ to near zero before the end of the recording.

Axis	Disturbance RMS	Residual RMS	Mean	Peak	DRR
Roll	6.150°	0.103°	−0.056°	0.391°	59.9×
Pitch	3.983°	0.513°	−0.006°	6.347°	7.8×

Table 3: Closed-loop attitude statistics over the full 165 s simulation interval. Disturbance RMS are included for reference.

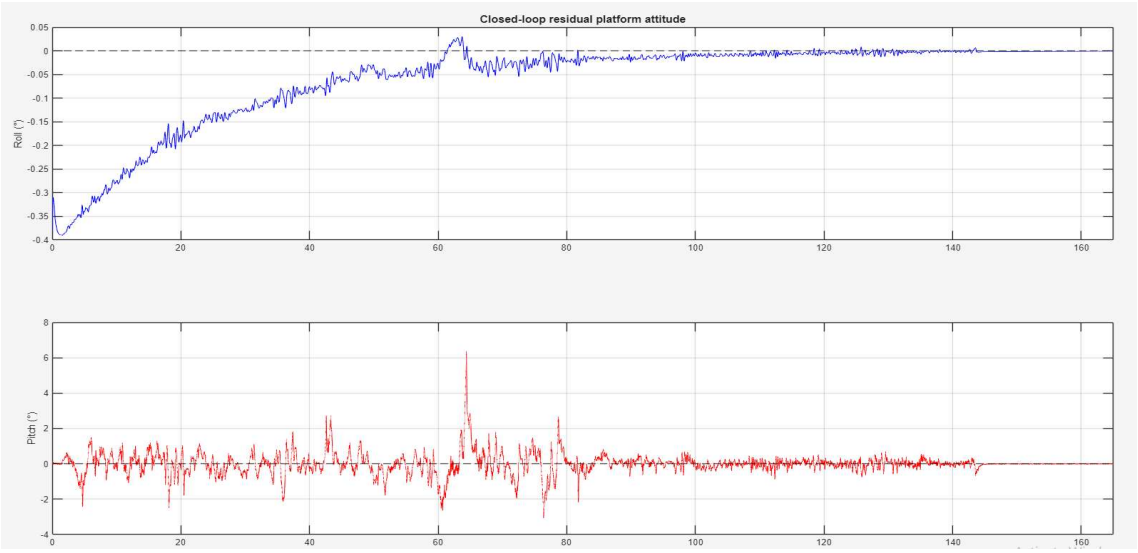


Figure 19: Closed-loop residual platform attitude over the acquired dataset. The roll axis, presented in the top image, exhibits gradual reduction of the initial bias during the simulation interval. The pitch axis, presented in the bottom image, exhibits the maximum residual response near $t = 65$ s.

The measured roll residual RMS of 0.103° remained close to the $\theta_{\mathbf{residual}} \leq 0.1°$ requirement later established in **RQ2**, while the pitch residual RMS was 0.513°. The measured roll residual RMS of 0.103° corresponds approximately to 9 cm projected vertical uncertainty at 50 m range according to Equation (13), while the pitch residual RMS of 0.513° corresponds to approximately 45 cm The frequency-domain analysis described in Section 3.2 characterises the spectral response of

the roll and pitch axes shown in Figure 20. The roll axis attains attenuation exceeding 40 dB across the entire 0.1 – 5 Hz band, with peak attenuation of 55.2 dB at 0.24 Hz corresponding to a power reduction of approximately 316×. The pitch axis exhibits a narrower effective bandwidth, with peak attenuation of 25.6 dB at DC, a 3 dB cut-off frequency of 1.07 Hz, and negative attenuation of approximately −5 dB across the 5 – 13 Hz band. The measured negative attenuation corresponds to amplification of disturbance energy within this frequency range.

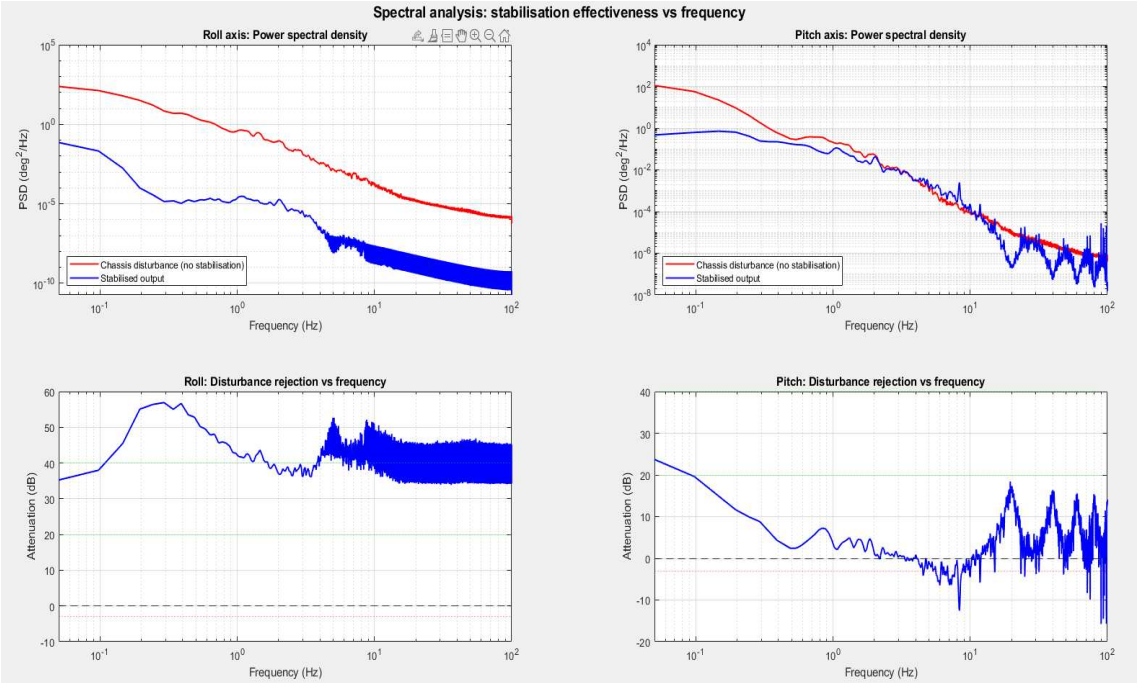


Figure 20: PSD illustrated in the top image and per-frequency attenuation illustrated in the bottom image for the roll axis and pitch axes, respectively. The roll axis exhibits attenuation exceeding 40 dB across the operational band. The pitch axis exhibits a 1.07 Hz cut-off frequency together with amplification across the across the 5-13 Hz band.

The combined simulation results corresponding to the research questions defined in Section 2. are presented as follows. **RQ1** evaluates the achievable RMS attenuation of terrain-induced roll and pitch disturbances. The closed-loop simulation, driven by recorded UGV chassis attitude data, produced a roll DRR of 59.9× (reducing 6.150° to 0.103° RMS) and a pitch DRR of 7.8× (reducing 3.983° to 0.513° RMS) under disturbances exceeding ±27°.

RQ2 evaluates the residual stabilisation error using the range-dependent LiDAR error budget. Equation (13) defines the relationship between θ_{residual} and projected vertical uncertainty. For the operational ranges evaluated in this work, the error budget shows that $\theta_{\text{residual}} \leq 0.1°$ is required to maintain $\sigma_{\text{total}} \leq 10\,\text{cm}$ at 50 m range. The simulated roll axis was $\theta_{\text{residual}} = 0.103°$, while the simulated pitch residual was $\theta_{\text{residual}} = 0.513°$. The stability thresholds described in Table 2 were evaluated using the detection mechanism described in Section 7.1.4. Over the full 165 s recording interval, the combined stability signal S_{stable} remained active for 61.27 % of the recording. The roll axis satisfied the stability criterion during 100% of the recording, while the pitch axis satisfied the stability criterion during 61.27 % of the recording. The longest continuous stable interval was 20.47 s, while the mean stable interval duration was 2.15 s. The aforementioned results regarding stability detection are shown in Figure 21, where the top subplot presents the stabilised platform attitude, where roll is represented in blue and pitch in red. The middle subplots show the corresponding binary stability signals for the roll and pitch axes. The bottom subplot illustrates the combined stability signal, indicating periods which LiDAR acquisition is permitted. The platform satisfied the combined stability criteria for 61.3 % of the recording duration.

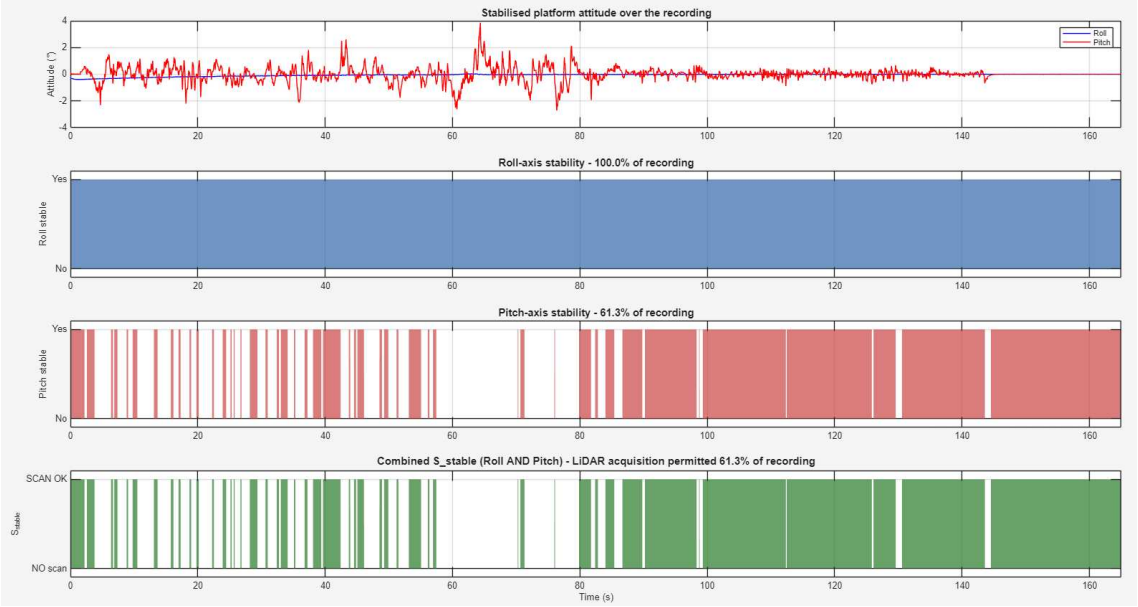


Figure 21: Stability detection results for the combined stability signal together with the individual roll and pitch stability signals over the 165 s simulation interval.

RQ3 evaluates the geometric impact of active stabilisation on reconstructed point cloud quality relative to the rigidly mounted baseline. The point cloud experiment described in Section 7.4.3 utilised a 4 s segment of the recorded chassis motion spanning $t = 50$ s to $t = 54$ s as the disturbance input. The window had roll RMS 4.48° and pitch RMS 5.28°. The corresponding platform residual across the same interval was roll RMS 0.05° and pitch RMS 0.62°. The three resulting clouds, shown in Figure 22, exhibit substantial vertical smearing in the rigidly mounted baseline cloud together with a centroid drift of approximately +17 cm, while the stabilised cloud exhibits a centroid offset of approximately −1.1 cm relative to the reference cloud. Quantitative comparison using ICP, C2C, and M3C2 is summarised in Table 4 and visualised in Figures 23 and 24. The distribution of cloud-to-cloud distances is shown in Figure 25. The stabilised configuration produced a 4.0× reduction in ICP registration RMSE, a 15.6× reduction in mean C2C distance, and a 15.7× reduction in significant M3C2 change rate relative to the rigidly mounted baseline. The proportion of points producing valid M3C2 measurements increased from 43.3 % in the rigidly mounted baseline cloud to 100 % in the stabilised cloud.

Metric	**Rigid mount**	**Stabilised**	**Ratio**
ICP RMSE	2.86 cm	0.71 cm	4.0×
C2C mean	10.78 cm	0.69 cm	15.6×
C2C median	10.03 cm	0.57 cm	17.6×
C2C 95th percentile	23.91 cm	1.81 cm	13.2×
M3C2 valid points	43.3%	100.0%	—
M3C2 significant change rate	43.9%	2.8%	15.7×

Table 4: Point cloud comparison metrics relative to the reference cloud for the rigidly mounted and actively stabilised configurations.

The signed-distance visualisation in Figure 24 shows substantially larger signed-distance variation in the rigidly mounted baseline cloud compared to the stabilised cloud. The results shown in Figures 23–25 quantify the geometric differences between the rigidly mounted baseline and actively stabilised sensor configurations evaluated in **RQ3**.

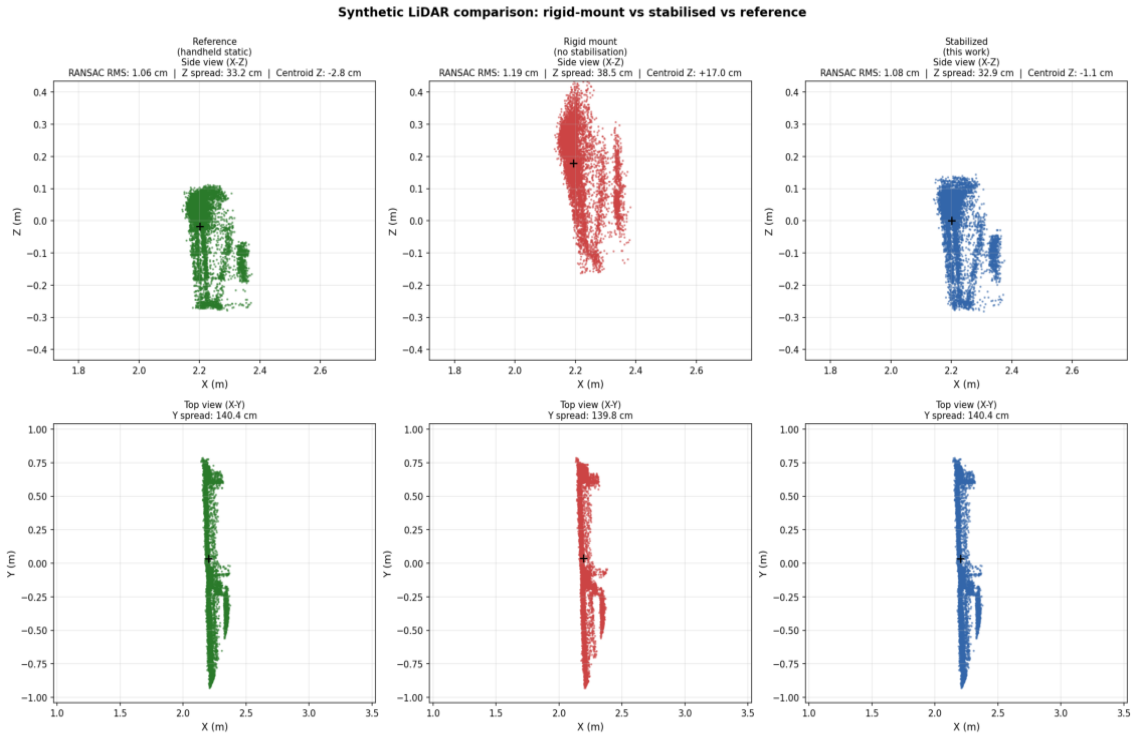


Figure 22: Synthetic point-cloud comparison using the bench reference. Left: handheld reference. Centre: rigidly mounted configuration under recorded UGV disturbance. Right: actively stabilised configuration using the simulated residual response. Top row: side view (X-Z). Bottom row: top view (X-Y).

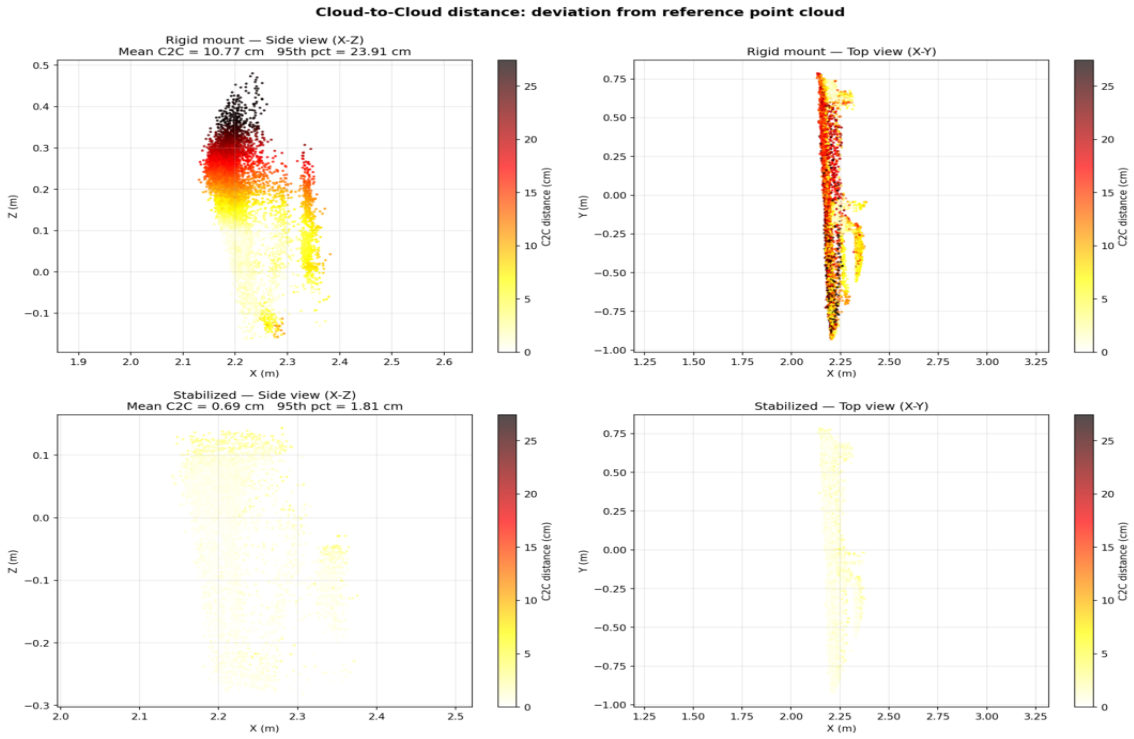


Figure 23: Spatial distribution of point-wise C2C distances relative to the static reference, colour-coded by magnitude. Top: rigidly mounted cloud. Bottom: stabilised cloud.

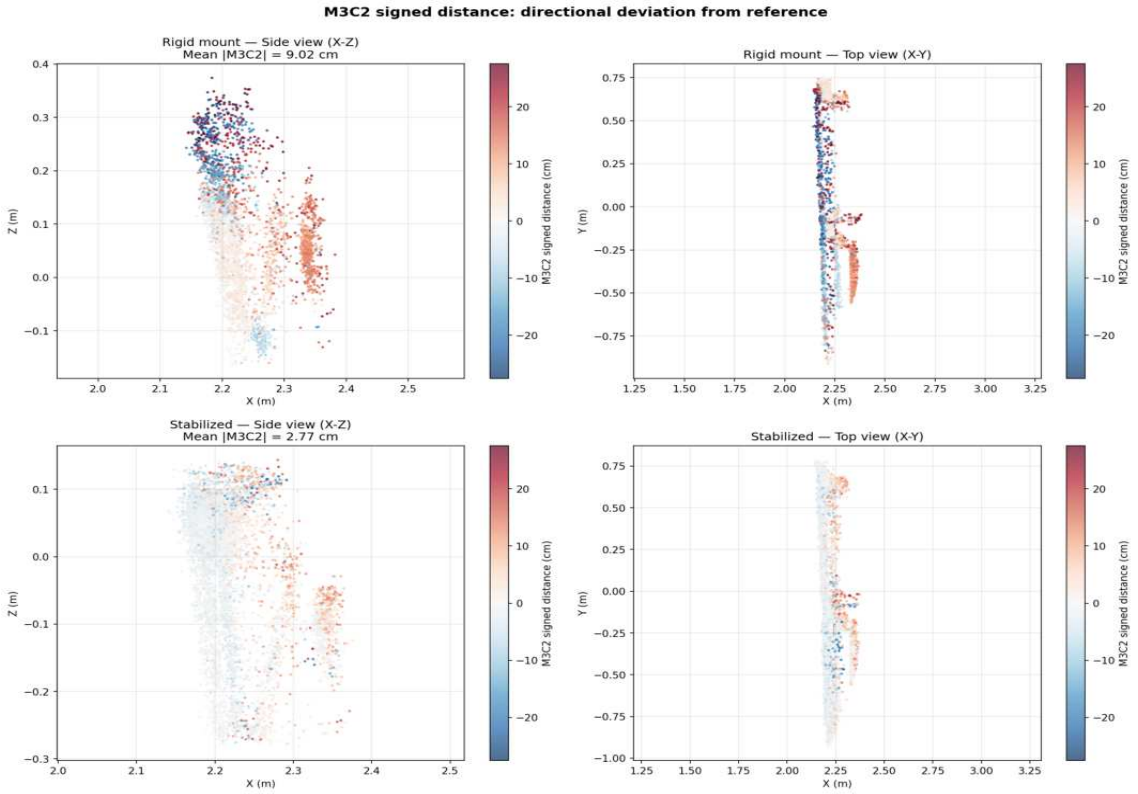


Figure 24: Spatial distribution of M3C2 signed distances relative to the static reference. The rigidly mounted cloud (top) exhibits substantially larger signed-distance variation compared to the stabilised cloud (bottom).

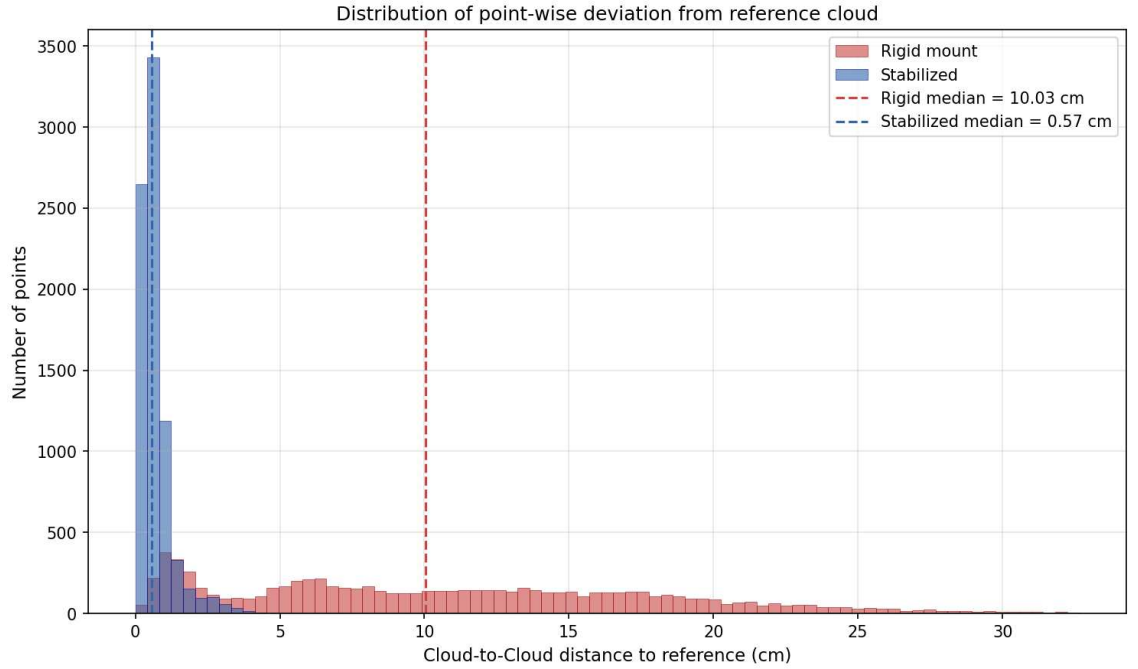
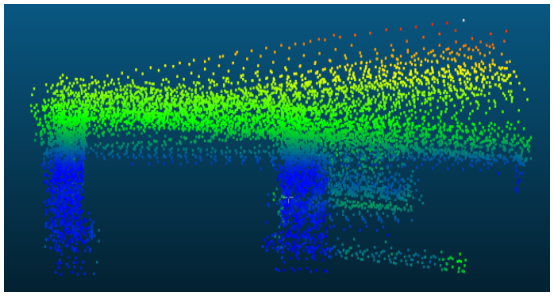
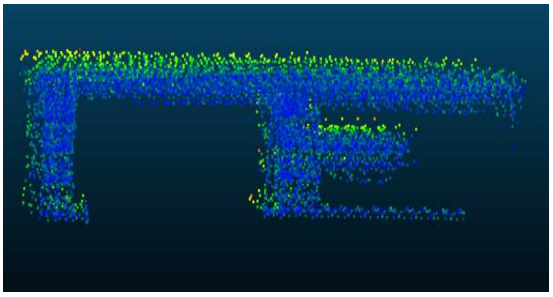


Figure 25: Distribution of point-wise cloud-to-cloud distances relative to the reference cloud. The stabilised distribution exhibits a median distance of 0.57 cm, while the rigidly mounted distribution exhibits a median distance of 10.03 cm together with point-wise deviations exceeding 25 cm.

In addition to the quantitative metrics presented in Table 4, a visual comparison of the acquired point clouds further illustrates the effect of active stabilisation. The rigidly mounted configuration, represented in Figure 26a, exhibited increased point dispersion and geometric distortion, whereas the stabilised configuration produced a more geometric consistent reconstruction of the reference, represented in Figure 26b. These observations are consistent with the reductions in ICP registration error, C2C distance, and M3C2 deviation reported in Table 4. Across all evaluated metrics, the actively stabilised configuration exhibited reduced residual attitude variation and improved point cloud quality relative to the rigidly mounted baseline.



(a) Rigidly Mounted Stabilisation Point Cloud



(b) Stabilised Point Cloud

Figure 26: Visual comparison between the rigidly mounted and actively stabilised LiDAR configurations relative to the static reference target represented in Figure 15b. The stabilised configuration exhibited reduced point spread and improved geometric consistency than the rigidly mounted case.

9. Discussion

The purpose of this thesis was to reduce long-range geometric distortion in LiDAR point clouds caused by terrain-induced UGV disturbances in uneven terrain. To address this problem, an active 2-DOF stabilisation controller was developed to minimise roll and pitch motion of the sensor platform. Rather than evaluating the controller solely in terms of angular attenuation, the presented framework linked residual platform motion to range-dependent spatial uncertainty and point cloud degradation. The discussion therefore focuses on how the stabilisation system influenced LiDAR geometric consistency and error, as well as the practical implications for autonomous UGV perception.

9.1 Disturbance Rejection and Stabilisation Performance

The results demonstrate that the developed stabilisation system substantially reduced terrain-induced roll and pitch motion of the LiDAR platform, where roll achieved a DRR of 59.9× and pitch achieved a DRR of 8× from the dataset acquired using the SLaRC UGV during field testing. The reduction in residual platform attitude directly improved the geometric consistency of the resulting synthetic point clouds, supporting the central motivation of this work that active stabilisation can preserve long-range LiDAR measurement integrity under dynamic conditions. The frequency-domain analysis further showed that the stabilisation system was most effective in attenuating low-frequency disturbances associated with large-scale terrain traversal and chassis inclination. Such behaviour is expected for a 2-DOF gimbal, where slower chassis attitude variations can be compensated more effectively than abrupt high-frequency shocks and vibration present in uneven terrain environments. A noticeable increase in residual attitude error was observed during the $60 - 80\,\mathrm{s}$ interval of the recording. The interval coincides with the largest pitch disturbances present in the dataset and represent a terrain section containing more severe excitation. The increased residual response suggests that the platform approached the limits of its disturbance rejection capability under these operating conditions. Pitch disturbances may be amplified through suspension dynamics or cross-axis interactions within the stabilisation mechanism, reducing the achievable attenuation compared to the roll axis.

The PSD-based attenuation analysis demonstrated that disturbance energy was significantly reduced across the low-frequency region most relevant to large-scale chassis inclination and terrain traversal. However, reduced attenuation and partial amplification were observed within approximately the 5–13 Hz frequency range. Such behaviour likely originates from a combination of limited closed-loop bandwidth and the mechanical assembly's cross-axis coupling. The presence of amplification in this frequency span suggests that the stabilisation platform behaves more as a coupled electromechanical structure with frequency-dependent dynamics rather than an ideal rigid-body system. Although these frequencies contributed less to large-scale geometric displacement than the low-frequency chassis motion, the results indicate that additional damping, structural stiffening, or more advanced control strategies may be required to suppress higher-frequency terrain-induced disturbances more effectively.

9.2 Range-Dependent LiDAR Error Behaviour

The uncertainty model defined in Equation 13 showed that the spatial effect of residual platform attitude error increased proportionally with sensing range. Residual platform errors below 1° nevertheless generated noticeable vertical displacement at longer ranges due to the range-dependent projection described in Equation 13. The synthetic point cloud comparisons in Figure 23 and Figure 24 reflected this behaviour, where the rigidly mounted baseline exhibited progressively larger geometric deviation relative to the stabilised configuration at longer distances.

Equation 14 further defined the range at which the angular projection term exceeded the measured LiDAR noise floor σ_{LiDAR}. Below R_{crit}, the total measurement uncertainty remained dominated primarily by intrinsic sensor noise characterised during the static reference outdoor gym experiment. The comparison results additionally showcased that the rigidly mounted configuration retained comparatively high geometric consistency at shorter ranges, where the spatial projection of attitude error remained limited. However, the geometric deviation introduced by platform motion

accumulated rapidly with increasing range, producing substantially larger geometric deviation in the unstabilised case. The stabilised point clouds consistently retained lower geometric deviation throughout both C2C and M3C2 algorithm analyses. Hence, the results indicate that stabilisation performance should not be solely evaluated in terms of angular attenuation, but rather in terms of its effect on spatial measurement consistency relevant to downstream perception tasks.

9.3 Experimental and System Limitations

The presented evaluation framework relied on synthetic point cloud generation rather than direct acquisition from a physically stabilised LiDAR platform during terrain traversal. The rigidly mounted and stabilised point clouds were generated by applying the recorded 165 s terrain traversal dataset and the simulated stabilisation controller, respectively, to the same static reference cloud described in Section 7.4.1. Similarly, the LiDAR noise floor σ_{LiDAR}, used in Equation 13, was obtained from the static reference experiment using RANSAC-based plane fitting. The presented framework, therefore, combined experimentally measured sensor behaviour from real uneven terrain conditions with controlled disturbance application, allowing repeatable comparison of the geometric effect of stabilisation under identical conditions.

Several limitations remain within the presented analysis. The Simscape Multibody model did not capture multiple non-ideal mechanical effects present in a physical implementation. These include structural flexibility, actuator backlash, cable interaction and twisting, and high-frequency vibration transmission. Such effects may introduce unintended platform motion and phase delay within the closed-loop response, particularly during continuous and rapid terrain excitation, where stabilisation accuracy becomes increasingly sensitive to non-ideal mechanical behaviour. The presented framework relied on Simscape Multibody rather than full robotic simulation environments such as Gazebo or O3DE. Direct integration of rigid-body dynamics and controller development was prioritised over environmental interaction and full autonomy-stack simulation, which were outside the scope of this thesis. Furthermore, the stabilised point clouds shown in Figure 23 and Figure 24 were not validated against measurements acquired from a fully integrated LiDAR platform during real terrain traversal. The presented results should therefore be interpreted primarily as an evaluation of the geometric influence of platform attitude stabilisation under representative terrain-induced disturbances. The framework provides a foundation for future physical implementation of a suitable stabilisation gimbal.

The stabilisation was implemented as a serial 2-DOF gimbal providing active roll and pitch compensation. This configuration targeted the dominant disturbances observed in the recorded UGV traversal dataset described in Section 7.4.1. Since the main focus of the thesis regarded vertical geometric error, yaw stabilisation was excluded from the architecture, as yaw primarily influences azimuth alignment and scan registration rather than vertical geometric error. A more mechanically complex multi-axis platform, such as a Stewart platform, could potentially provide improved disturbance isolation and full six-degree-of-freedom stabilisation. However, the presented work specifically targeted terrain-induced roll and pitch disturbances responsible for the dominant vertical geometric error, making the simpler 2-DOF architecture more suitable for the intended evaluation framework. The dual-IMU configuration enabled direct evaluation of disturbance rejection performance by separating chassis disturbance input from residual platform motion throughout the recorded terrain traversal. Additionally, absolute encoder feedback may influence stabilisation precision, since small angular estimation errors directly propagate into residual platform attitude error.

9.4 Practical Implications for UGV Perception

Applications such as localisation, mapping, terrain reconstruction, obstacle detection, and SLAM rely on geometrically consistent point cloud measurements between successive sensor observations. Residual platform motion introduces spatial distortion that may reduce registration accuracy, increase localisation drift, and degrade reconstruction quality, especially in uneven terrain environments. The presented results, therefore, indicate that active platform stabilisation can improve the reliability of long-range LiDAR-based perception for autonomous off-road UGV systems.

10. Conclusions

This thesis investigated the use of active 2-DOF stabilisation to reduce terrain-induced geometric distortion in long-range LiDAR perception for UGV systems operating in uneven terrain environments, such as forests and obstructed areas. A serial roll and pitch stabilised platform was developed and evaluated using experimentally recorded attitude data acquired from the SLaRC UGV and the 3DM-GV7-AHRS IMU, combined with a Simscape Multibody stabilisation controller and a synthetic point cloud evaluation framework. Rather than evaluating stabilisation performance solely through angular disturbance rejection, the presented methodology linked residual platform attitude error to range-dependent LiDAR geometric deviation and point cloud consistency. The work, therefore, addressed both stabilisation performance and its practical implications for long-range perception quality during continuous terrain traversal.

RQ1 investigated the level of terrain-induced roll and pitch attenuation achievable using active 2-DOF stabilisation. The developed stabilisation system substantially reduced residual platform motion during the recorded UGV traversal dataset, where roll achieved a DRR of 59.9× and pitch achieved a DRR of 8× relative to the rigidly mounted baseline on the SLaRC UGV. The PSD analysis further showed that the controller was most effective against low-frequency terrain-induced disturbances, while reduced attenuation was observed at higher frequencies.

RQ2 investigated the level of residual stabilisation accuracy required to maintain an acceptable range-dependent LiDAR geometric deviation during continuous UGV motion. The developed LiDAR error budget demonstrated that residual platform attitude error must remain below approximately 0.1° to maintain total vertical uncertainty below 10 cm at a sensing range of 50 m. The stabilised roll axis achieved a residual error of 0.103°, while the pitch axis retained a larger residual error of 0.513° throughout the evaluated terrain dataset. The implemented stability detection framework further showed that the combined stabilisation signal remained within the defined stability thresholds during 61.27 % of the dataset, with the roll axis remaining stable throughout the complete dataset. The results, therefore, demonstrated that long-range geometric consistency becomes highly sensitive to even small residual platform attitude errors during continuous uneven terrain traversal.

RQ3 investigated whether active 2-DOF stabilisation improves LiDAR geometric consistency relative to a rigidly mounted baseline. The synthetic point cloud comparison framework presented in Figure 23 and Figure 24 demonstrated that the stabilised configuration consistently retained lower geometric deviation throughout both C2C and M3C2 analyses, where the rigidly mounted baseline exhibited substantially larger geometric distortion. The presented results, therefore, demonstrated that active roll and pitch stabilisation can measurably improve geometric consistency in long-range LiDAR perception in uneven terrain environments.

The presented work demonstrated that active platform stabilisation can measurably improve long-range LiDAR geometric consistency during uneven terrain UGV traversal. By directly linking residual platform attitude error to range-dependent spatial uncertainty and point cloud degradation, this thesis established a framework for evaluating stabilised perception systems beyond conventional angular disturbance rejection metrics. The results further indicate that active stabilisation may improve the reliability of downstream perception tasks such as localisation, mapping, obstacle detection, and SLAM in autonomous off-road environments.

11. Future Work

Although the proposed stabilisation framework demonstrated significant improvements in LiDAR geometric consistency under terrain-induced disturbances, several aspects remain for future investigation and development.

A primary continuation of this work involves the full physical implementation and experimental validation of the proposed stabilisation platform on the SLaRC UGV. While this thesis focused on the simulation-based evaluation using experimentally recorded terrain traversal data, real-world and real-time validation would enable evaluation of actuator dynamics, structural validation, latency, and hardware-related disturbances not fully representable in a simulation environment. In particular, paired experimental traversals using both rigidly mounted and actively stabilised configurations over identical terrain profiles would enable direct and controlled quantitive comparison using the developed, yet unused, temporal point cloud analysis framework.

The simulation results further indicate that the pitch axis exhibited lower disturbance rejection compared to the roll axis. Future work could therefore investigate mechanical redesign of the pitch-axis assembly to improve the disturbance rejection of this particular axis, specifically at higher frequency ranges. Furthermore, the mechanical platform was designed specifically around the evaluated Livox MID-360 LiDAR payload and therefore provides limited flexibility for integration of alternative or additional sensors. Future work may therefore investigate more modular platform architectures capable of accommodating larger and longer-range high-resolution LiDAR sensors,[2] where residual angular stabilisation error becomes increasingly significant due to the extended sensing range.

Although this thesis focused exclusively on roll and pitch stabilisation due to their dominant contribution to gravity-referenced LiDAR geometric distortion. Future work could additionally investigate the quantitative impact of active yaw stabilisation and more advanced multi-axis configurations, such as 6-DOF platforms, to assess the practical benefits and use cases across different domains and disturbance scenarios. Furthermore, more advanced sensing architectures, including dual-IMU sensor fusion approaches, may improve robustness against vibration-induced estimation uncertainty or sensor bias. Additionally, utilising absolute encoders on the sensor payload could also prove to be beneficial and remains an unexplored territory in stabilisation techniques for precise LiDAR measurements.

Finally, the proposed evaluation framework may be generalised to other adjacent domains, including aerial, maritime, and larger ground-based robotic platforms, where stabilised perception systems similarly operate under dynamic motion disturbance conditions.

[2] For example, the Ouster OS2 long-range LiDAR series: https://ouster.com/products/hardware/os2-lidar-sensor

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A WBS

The WBS, as well as all of the phases and tasks in the project is shown in Figure 27.

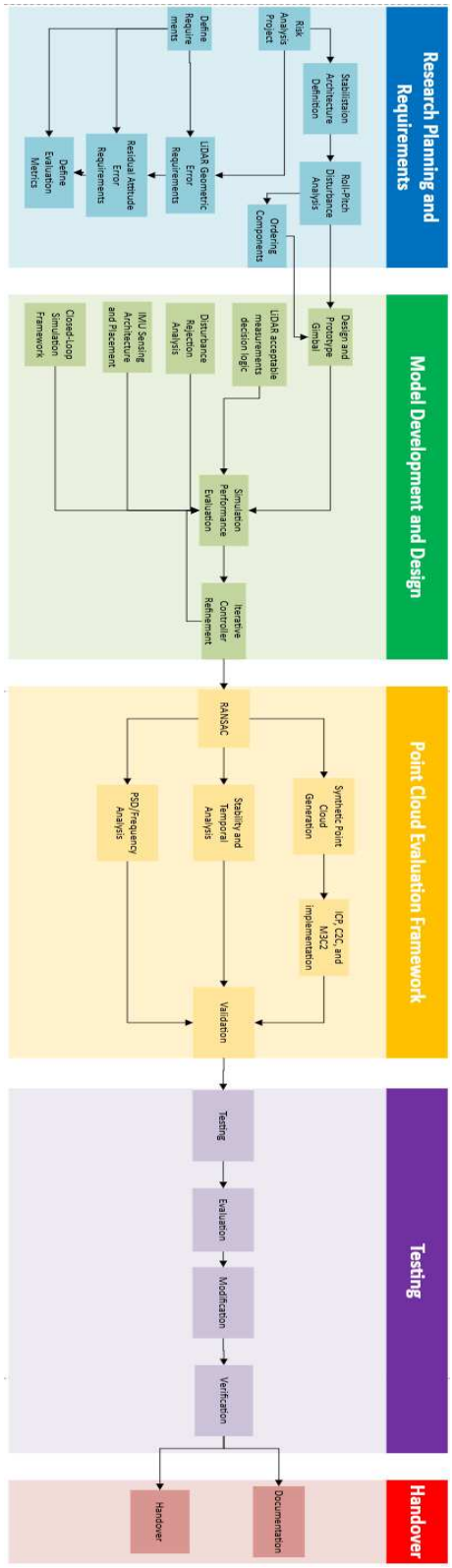


Figure 27: Work Breakdown Structure

B Risk Analysis

An overview of different risks that could arise during different phases of the project and as a whole is shown in Figure 28.

Risk Analysis for the Project

Conesqences Rating Guidelines:
1 - No/negligible consequences.
2 - Minor or no time/cost delays.
3 - Moderate time delays and/or cost concerns.
4 - Significant time and/or cost delays, with minor to moderate quality concerns.
5 - Project risks failure and/or significant quliaty concerns.

Score	Risk Level
0 -> 8	Low
9 -> 18	Medium
19 -> 25	High

ID	Title	Description	Date	Probability	Consequences	Risk level	Mitigation measure	Deadline
Risk 01	Mechanical Integration Complexity	Platform integration required several CAD iterations due to limited playload space and actuator placement constraints	2026-01-22	3	4	12	Simplified mehcanical design and iterative CAD refinement integration complexity	2026-05-20
Risk 02	Actuator Performance Limitations	Selected actuators may provide insufficient torque, bandwidth, or respons speed for effective terrain disturbance rejection.	2026-01-22	3	4	12	Perform preliminary torque estimation, simplify payload design, and validate actuator performance in simulation.	2026-05-20
Risk 03	IMU Measurement Uncertainty	IMU noise, drift, vibration sensitivity, or sensor latency may reduce stabilisation accuracy.	2026-01-22	4	3	12	Use fused INS-class oreintation estimation, sensor filtering, and deliberate IMU placement.	2026-05-20
Risk 04	Control Design and Tuning Challenges	Closed-loop controller tuning may lead to instability, oscillations, or insufficient disturbance rejection.	2026-01-22	3	3	9	Begin with simplified controller structures and incremenetally tune parameters in simulation.	2026-05-20
Risk 05	Experiemntal Repeatability Limitations	Real terrain traversal experiments may be difficult to repeat consistently under identical conditions.	2026-01-22	4	2	8	Record terrain traversal datasets and use replay-based evaluation fo	2026-05-20
Risk 06	Project Scope Overextension	The project scope may exceed available tiome due to simultaneous mechanical, control, simulation and perception objectives.	2026-01-22	3	5	15	Prioritise proof-of-concept stabilisation and reduce scope if necessary.	2026-05-20
Risk 07	Unclear Evaluation Metrics	Stabilisation performance metrics may initially be insufficiently defined or difficult to interpret quantatively.	2026-01-22	2	4	8	Define RMS, DRR, and point cloud evaluation metrics early in the project.	2026-05-20
Risk 08	Simulation-Reality Mismatch	Simulation results may not fully represent hardware behaviour or structural vibration effects.	2026-01-22	3	3	9	Use experimentally recorded terrain data and realistic actuator modelling where possible.	2026-05-20
Risk 09	Point Cloud Evaluation Complexity	Development of point cloud evaluation pipelines may require more implementation effort and complexity than anticipated	2026-01-22	3	3	9	Start evaluation framework development early and validate method	2026-05-20
Risk 10	Lead Time Delays	The actuator system lacks passive braking	2026-01-22	2	4	8	Order necessary components at earliest possibility	2026-05-20

Figure 28: Risk analysis of the project.

C Gantt Chart

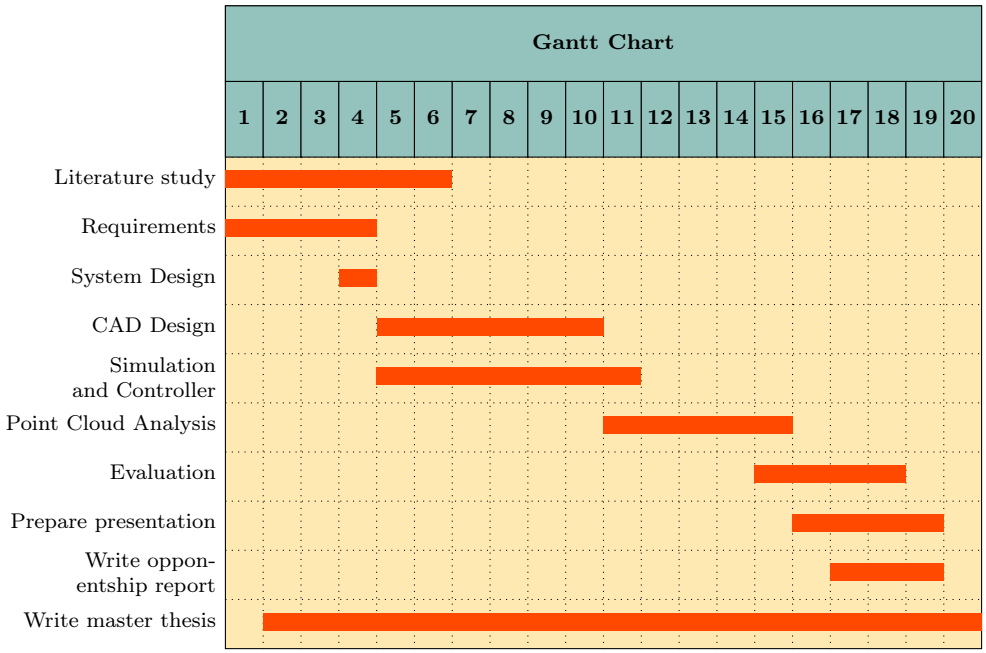


Figure 29: A time plan showing the main project phases over 20 weeks.

D Plant

The complete visualisation of the Plant discussed and implemented in Section 7.1.1

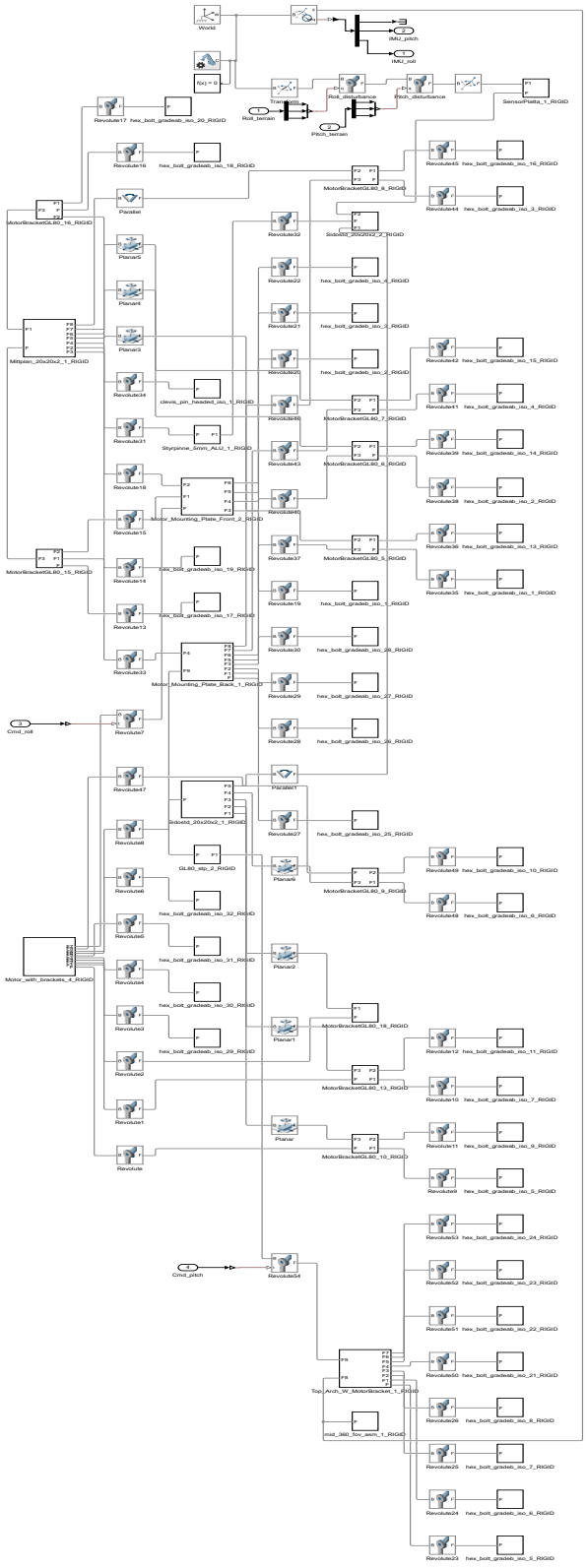


Figure 30: The complete visualisation of the Plant within Simscape Multibody.